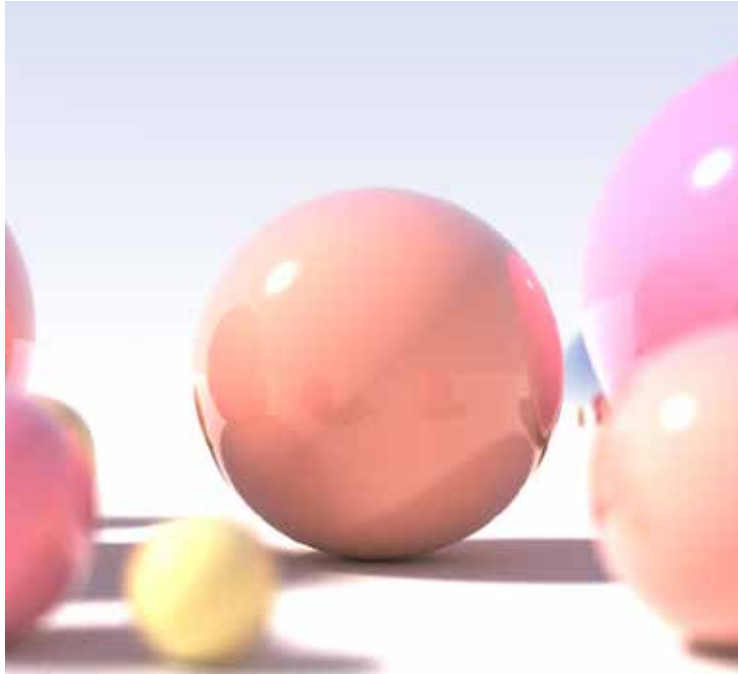


Ray Tracing

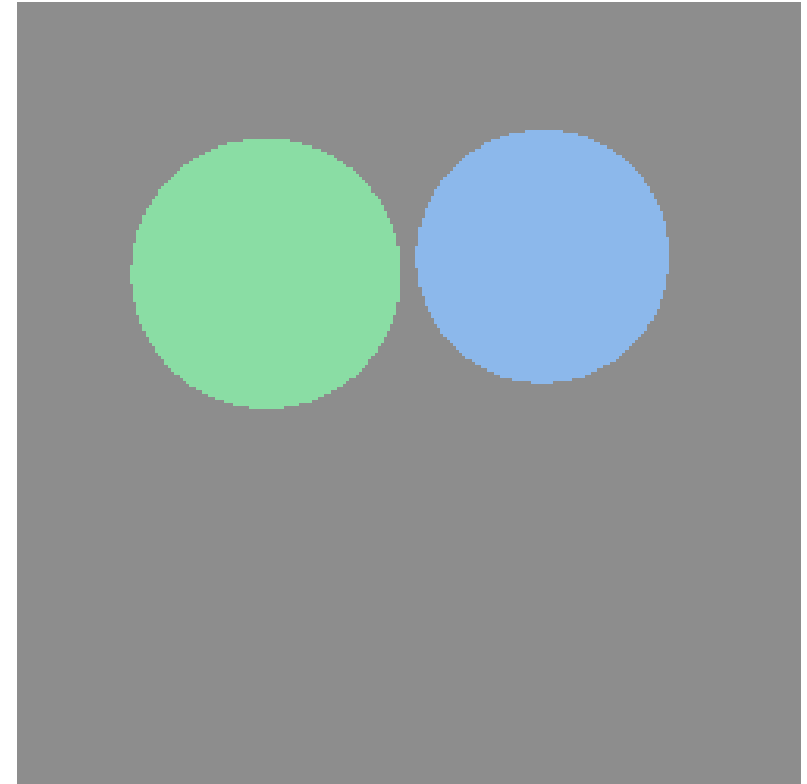
...not to be confused with Raytray Singh



Some Slides/Images adapted from Marschner and Shirley and David Levin

Ray Casting

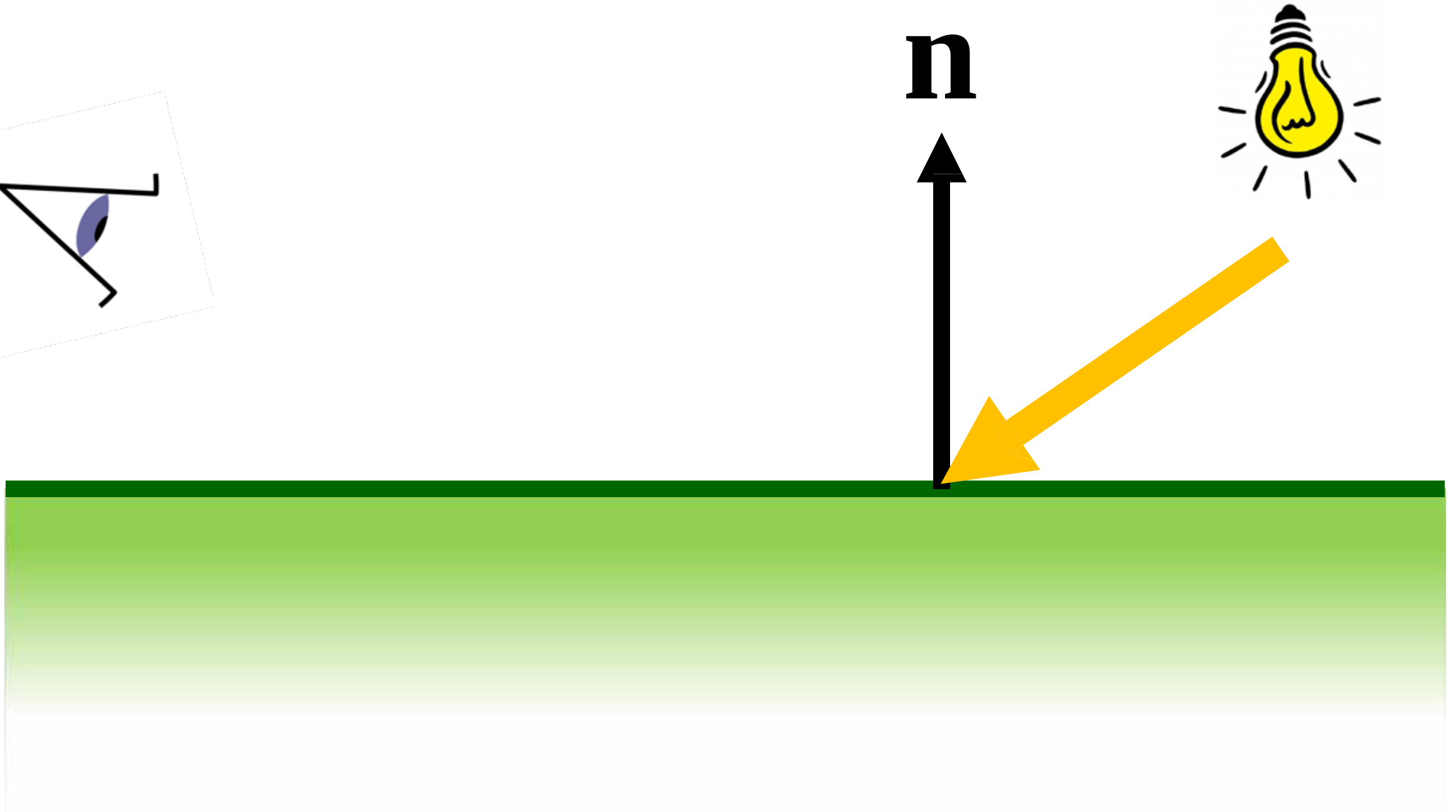
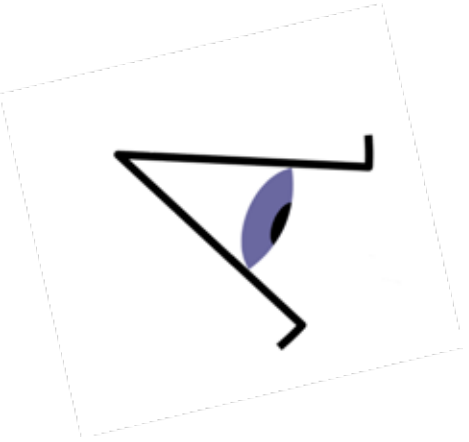
```
for iy in range(ny):  
    for ix in range(nx):  
        ray = camera.getRay(ix, iy)  
        firstSurface = scene.intersect(result, ray)  
        if firstSurface:  
            image.set(ix, iy, firstSurface.color)  
        else:  
            image.set(ix, iy, background.color)
```



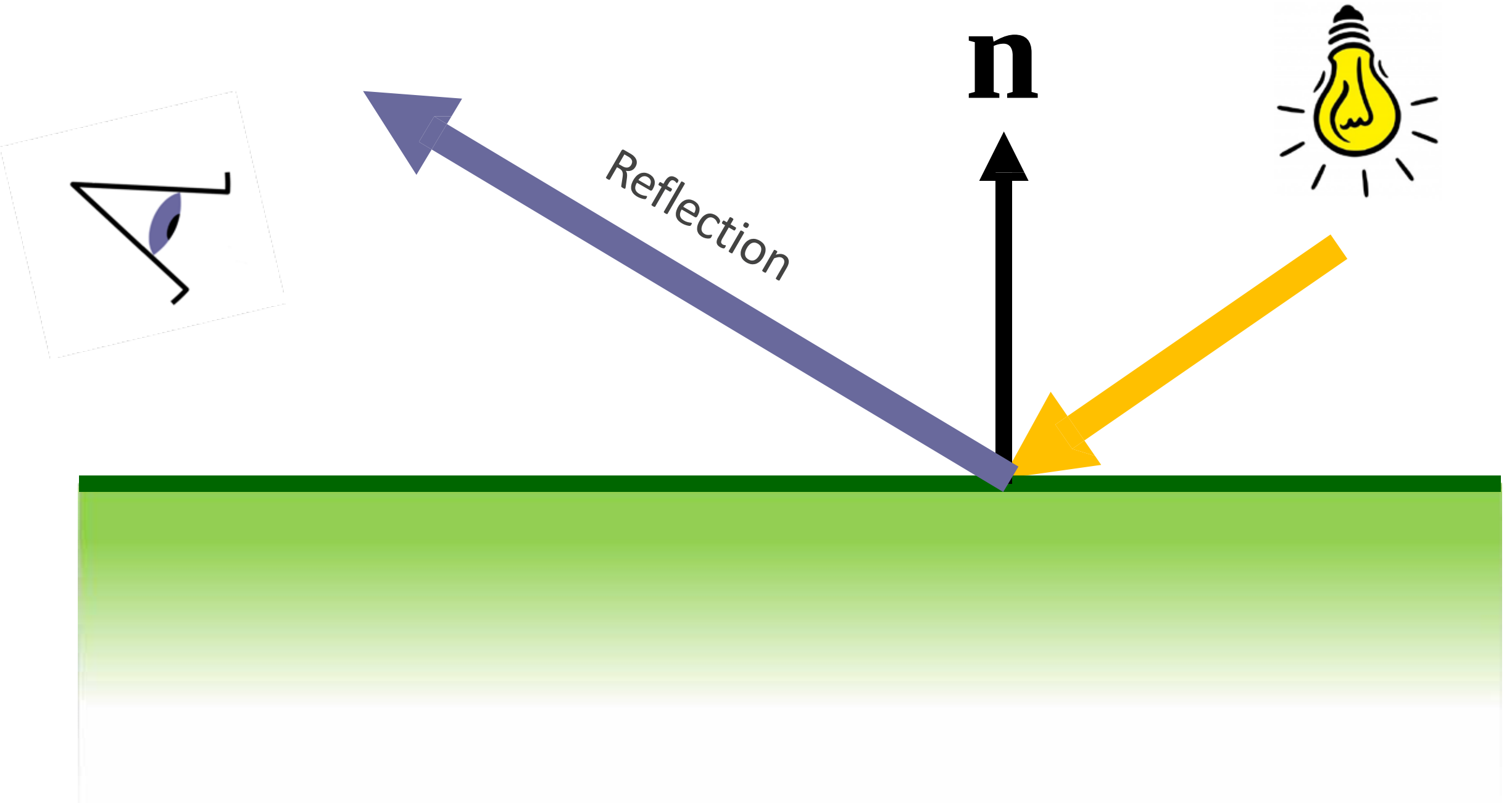
Light and Surfaces



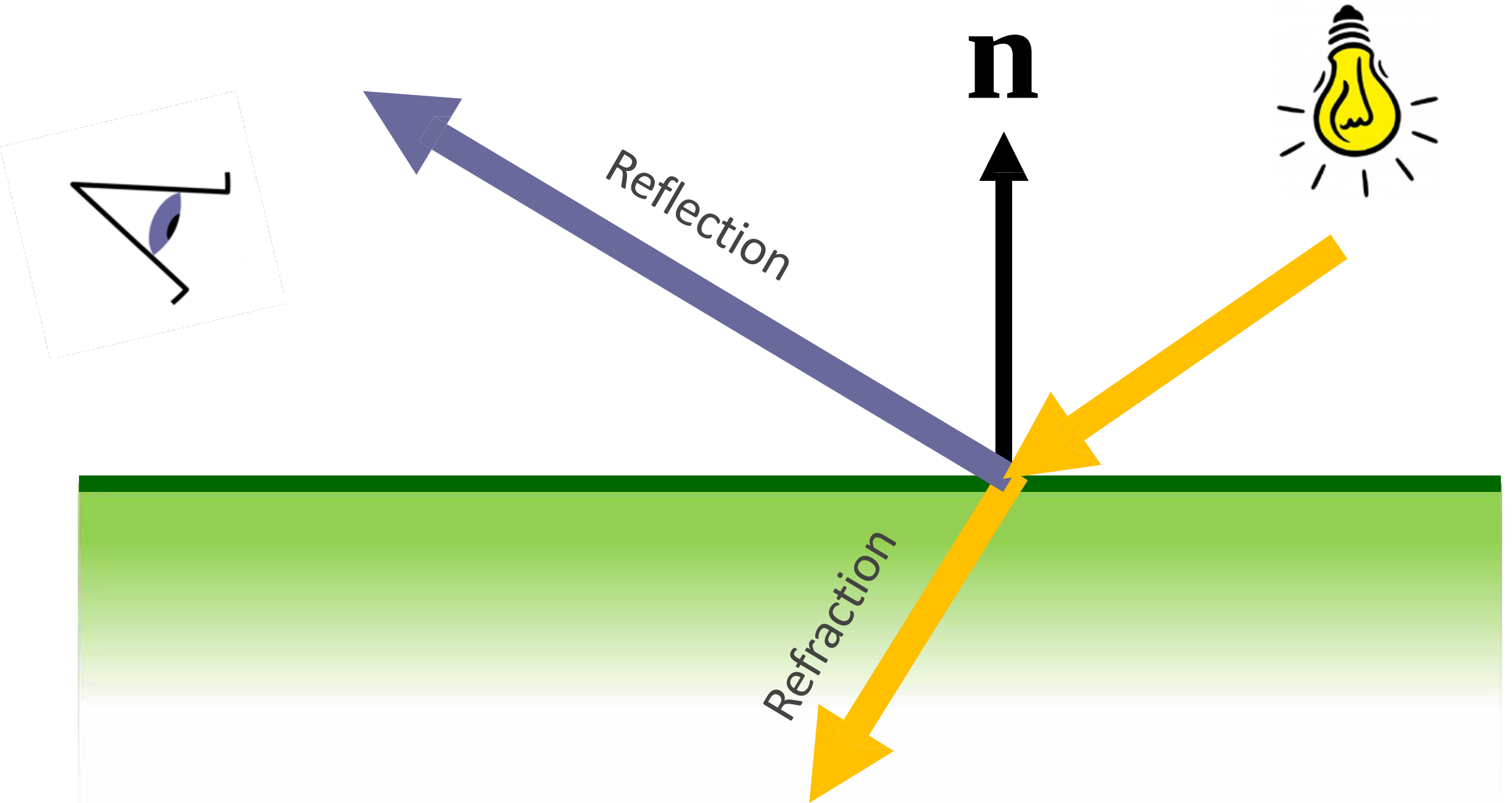
Light and Surfaces



Light and Surfaces

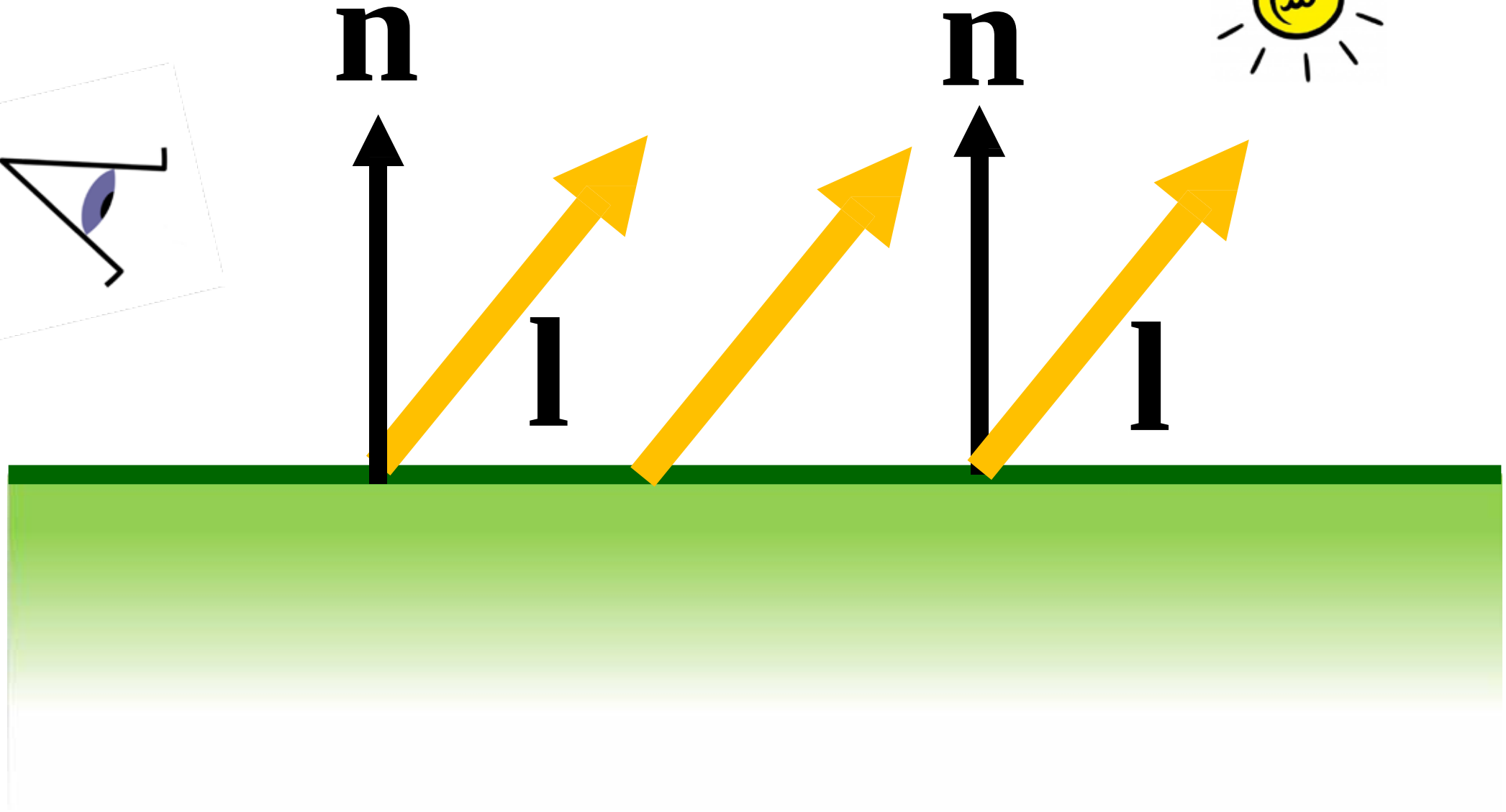
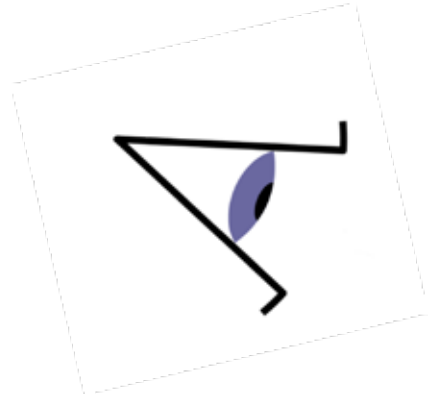


Light and Surfaces



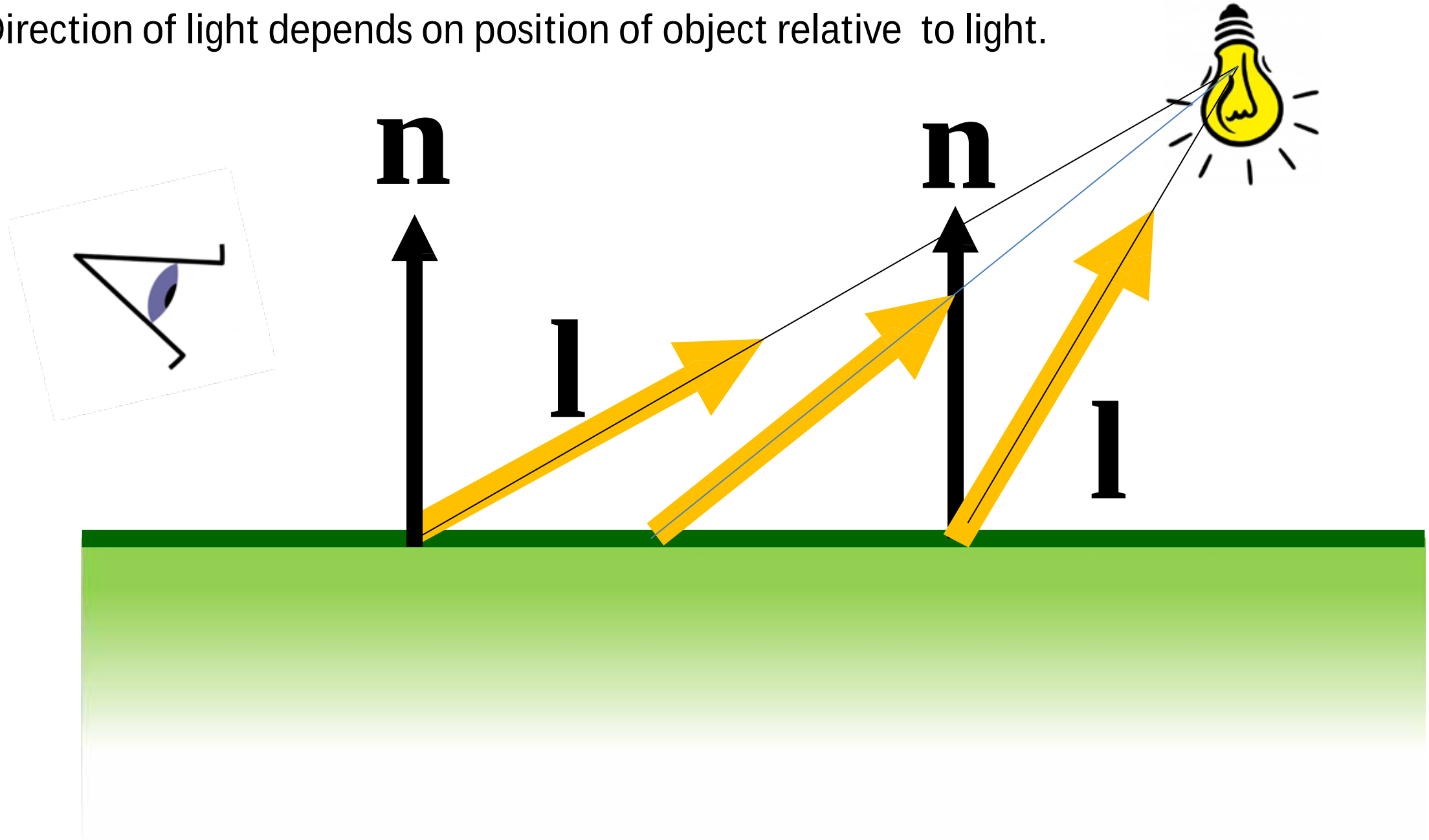
Directional Light

Direction of light is independent of the object. **Light is very far away**



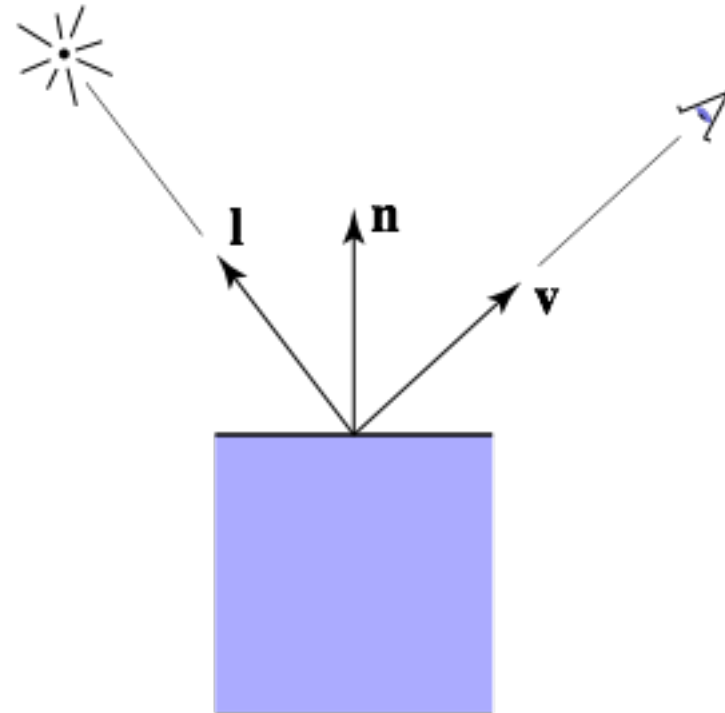
Point Light

Direction of light depends on position of object relative to light.



Shading

- Compute light reflected toward camera
- Inputs:
 - eye direction
 - light direction
(for each of many lights)
 - surface normal
 - surface parameters
(color, shininess, ...)

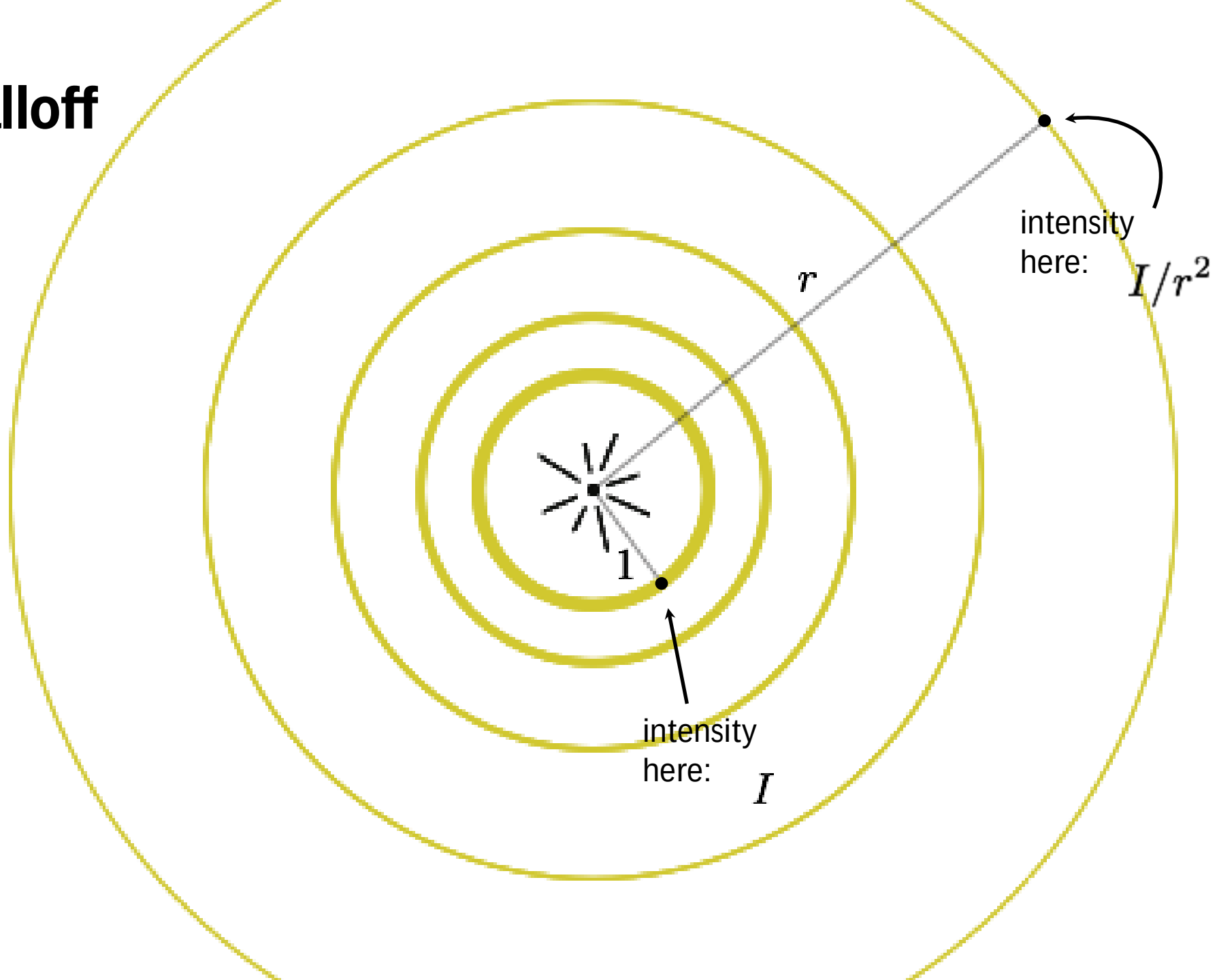


Computing the Normal at a Hit Point

- Polygon normal: cross product of two non-collinear edges.
- Implicit surface normal $f(p)=0$:
 $\text{gradient}(f)(p)$.
- Explicit parametric surface $f(a,b)$:

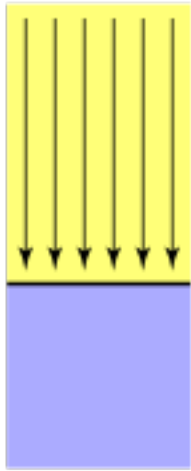
$$\delta f(s,b) / \delta s \times \delta f(a,t) / \delta t.$$

Light falloff

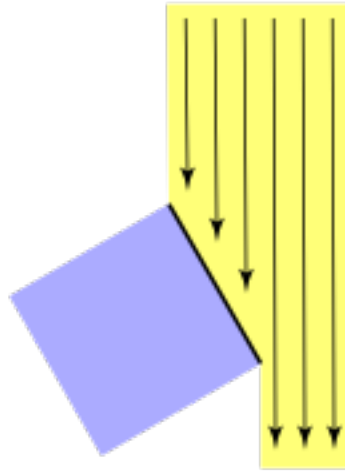


Diffuse reflection

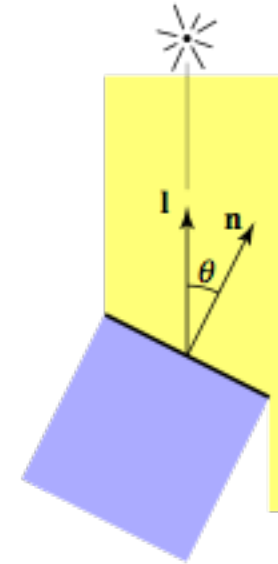
- Light is scattered uniformly in all directions
 - the surface color is the same for all viewing directions
- Lambert's cosine law



Top face of cube
receives a certain
amount of light



Top face of
60° rotated cube
intercepts half the light

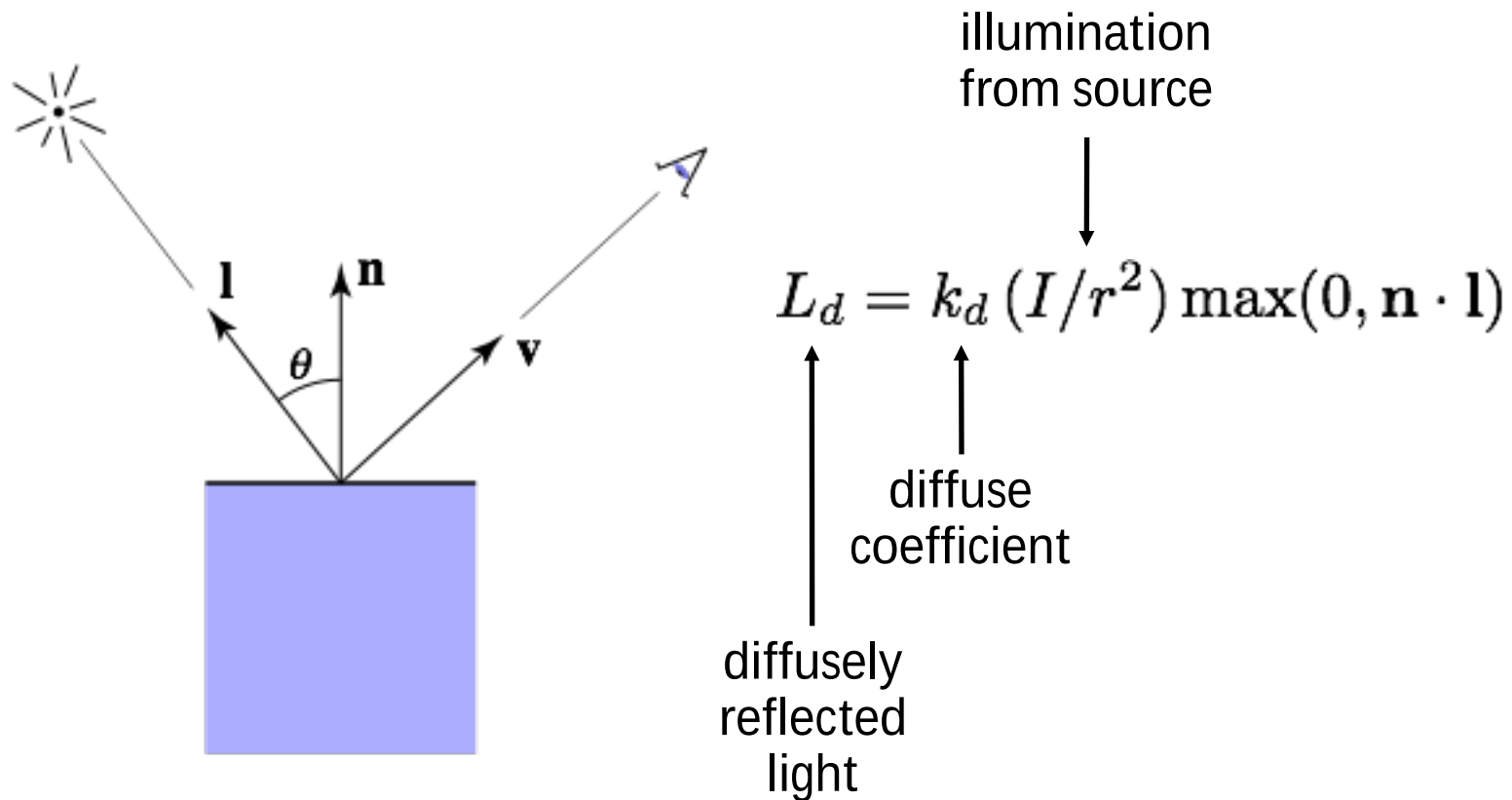


In general, light per unit
area is proportional to

$$\cos \theta = \mathbf{l} \cdot \mathbf{n}$$

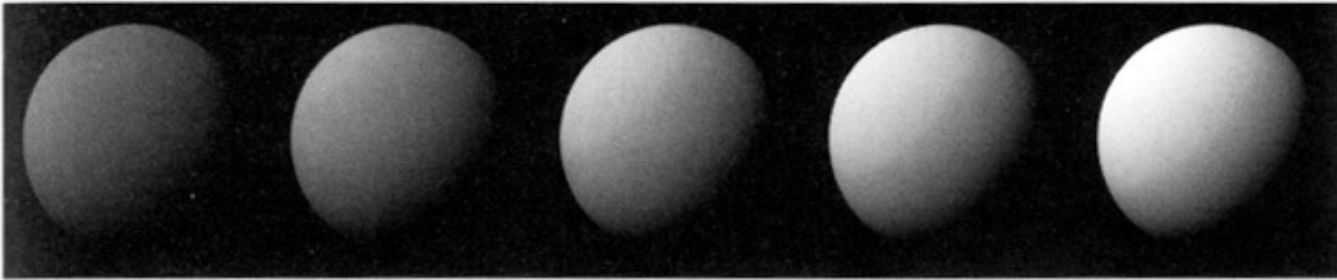
Lambertian shading

Shading independent of view direction



Lambertian shading

Produces a matte appearance



$k_d \longrightarrow$

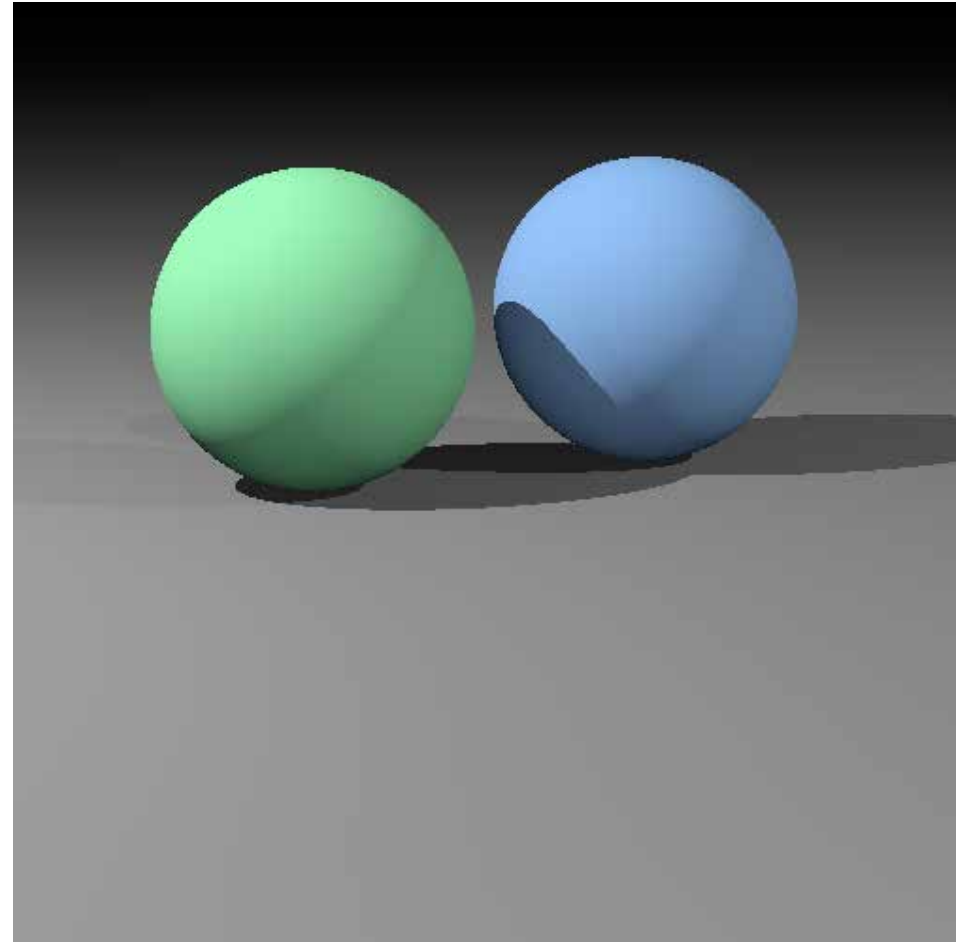


Image without shading

```
for iy in range(ny):  
    for ix in range(nx):  
        ray = camera.getRay(ix, iy)  
        firstSurface = scene.intersect(result, ray)  
        if firstSurface:  
            image.set(ix, iy, firstSurface.color)  
        else:  
            image.set(ix, iy, background.color)
```

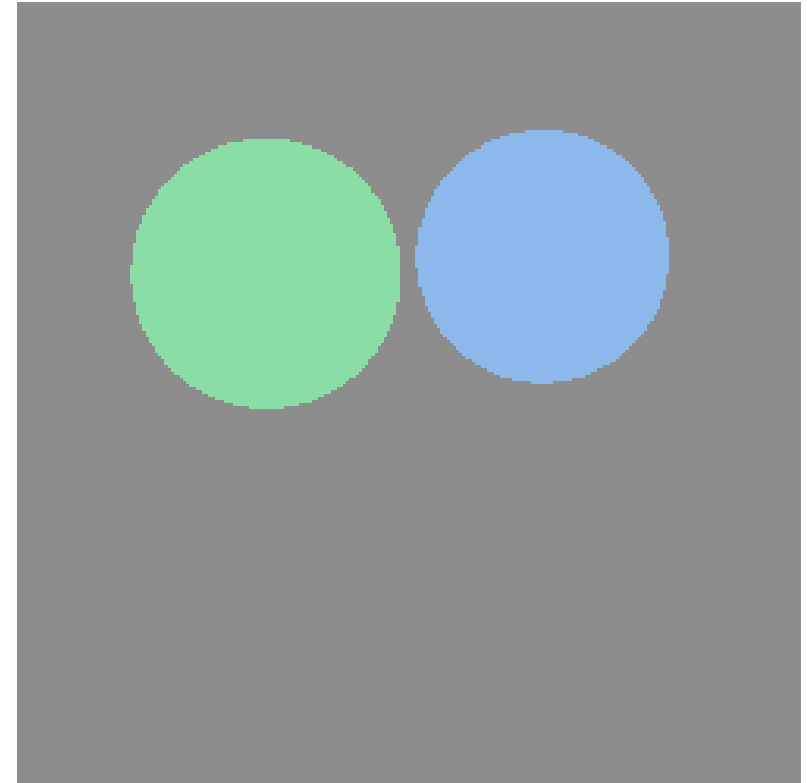
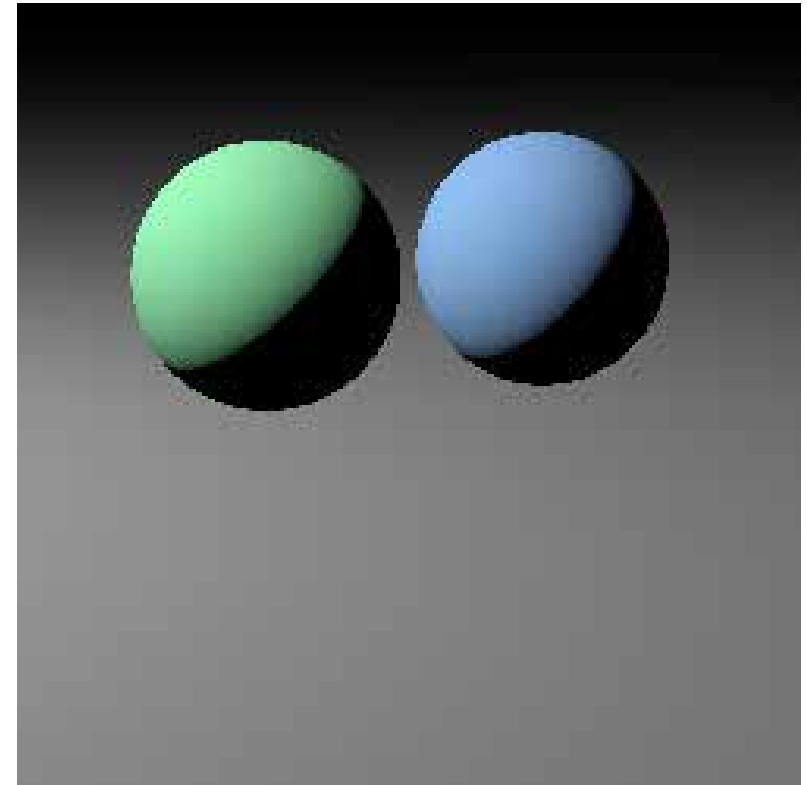


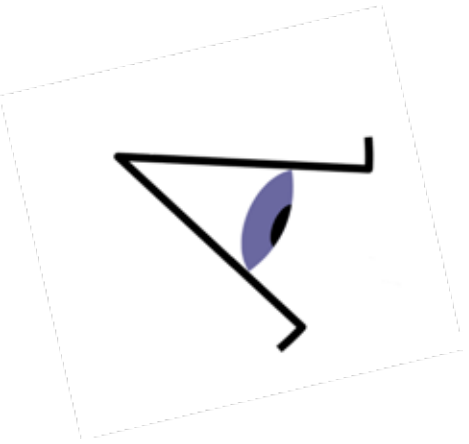
Image with shading

```
for iy in range(ny):
    for ix in range(nx):
        ray = camera.getRay(ix, iy)
        firstSurface = scene.intersect(result, ray)
        if firstSurface:
            image.set(ix, iy,
                        firstSurface.shade(ray,light,
                                            result.point,
                                            result.normal))
        else:
            image.set(ix, iy, background.color)

def surface.shade(ray,light,point,normal) :
    l=light.pos-position
    it= surface.k*light.intensity*max(0,normal.l)
    return surface.color*it
```



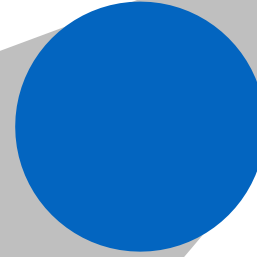
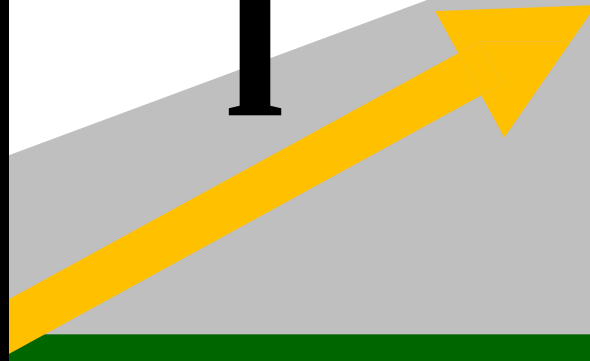
Adding Shadows



n



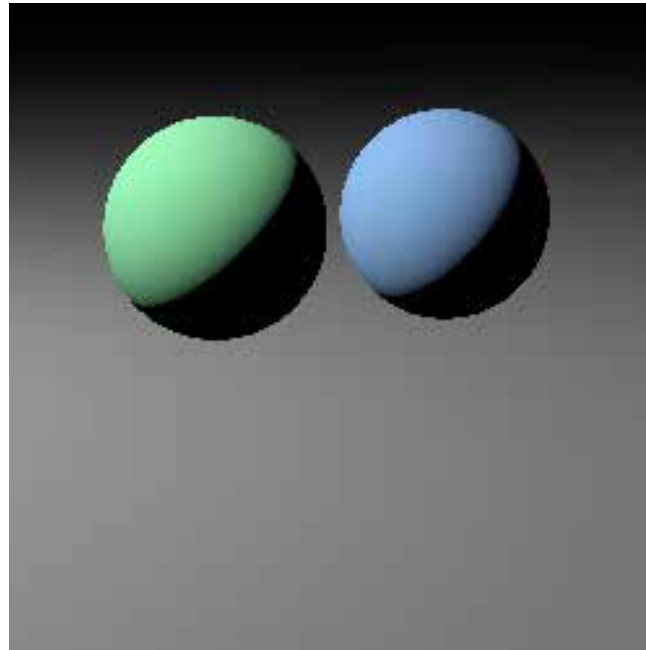
l



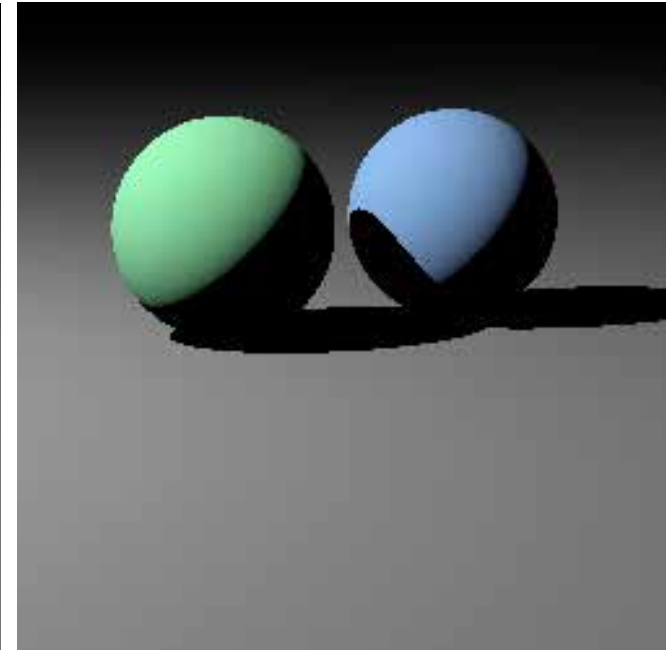
Shadows

- Surface is only illuminated if nothing blocks its view of the light.
- With ray tracing it's easy to check if a point in the scene is in shadow.
just shoot a ray from the point to the light and intersect it with the scene!

```
def surface.shade(ray,light,point,normal) :  
    l=light.pos-position  
    shadowray=(point,l)  
    if !scene.intersect(result,shadowray) :  
        it= surface.k*light.intensity*max(0,normal.l)  
        return surface.color*it  
    else:  
        return black
```



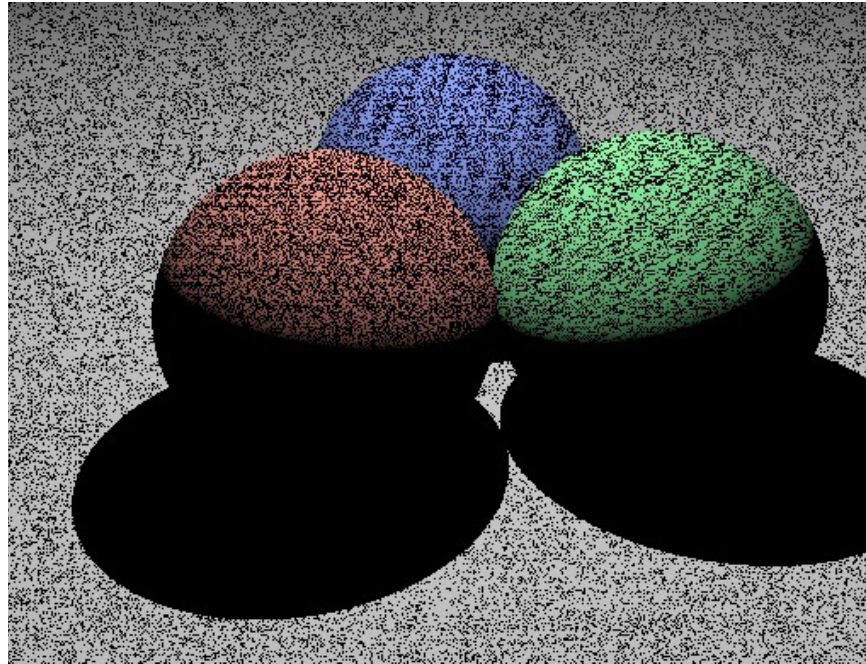
Without shadows



With shadows

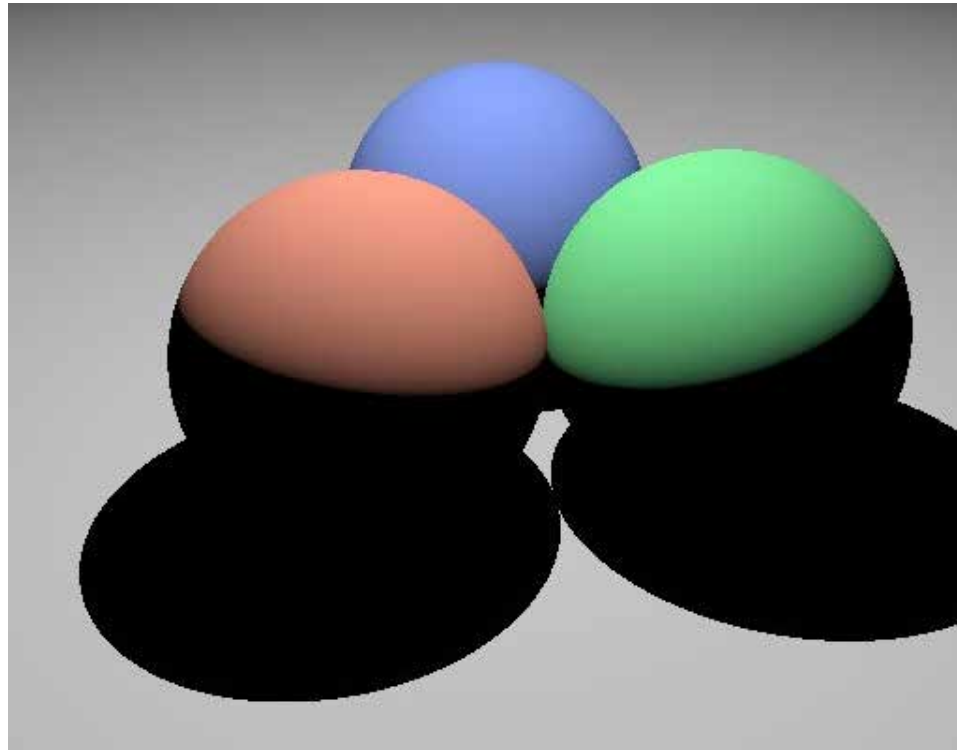
Classic shadow error

What's going on?



Classic shadow error

Start shadow rays just outside surface



Multiple lights

- Important to fill in black shadows
- Just loop over lights, add contributions
- Ambient shading
 - black shadows are not really right
 - one solution: dim light at camera
 - alternative: add a constant “ambient” color to the shading...

Ambient shading

Shading that does not depend on anything

- add constant color to account for disregarded illumination and fill in black shadows

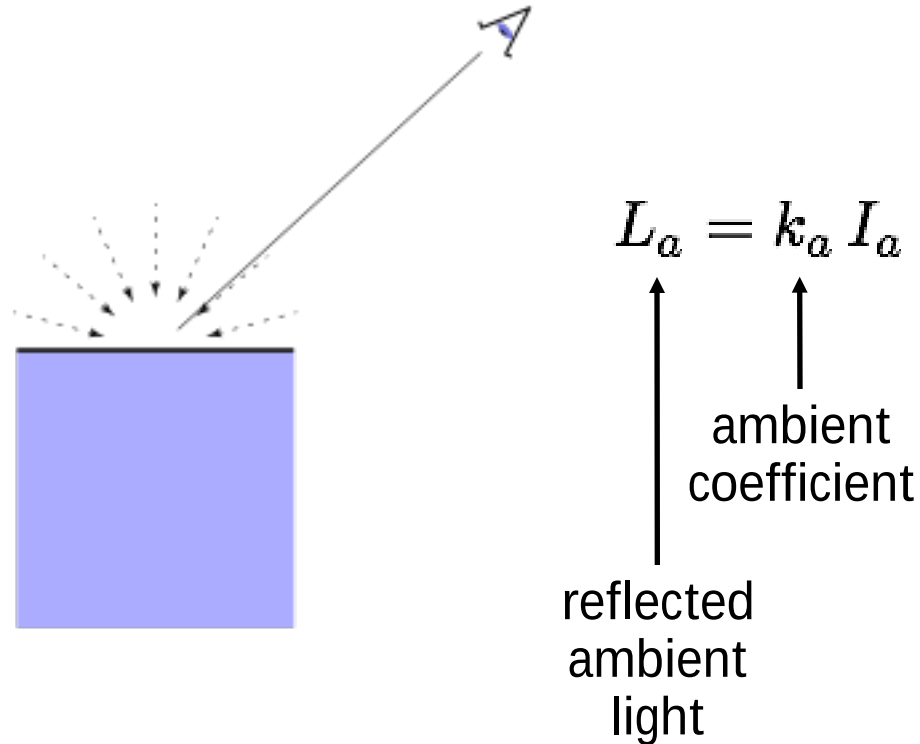
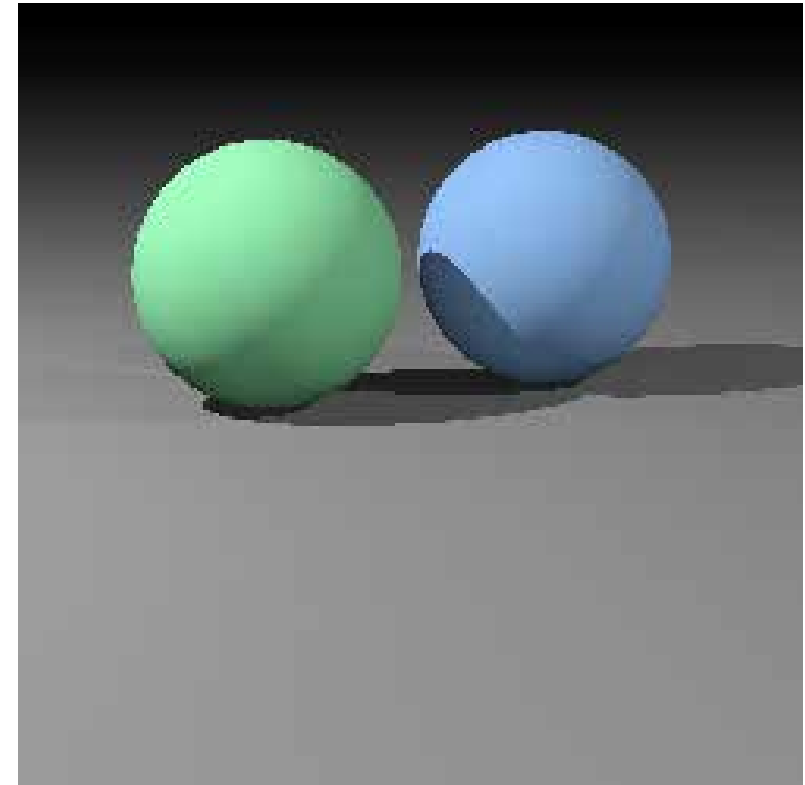


Image with multiple lights

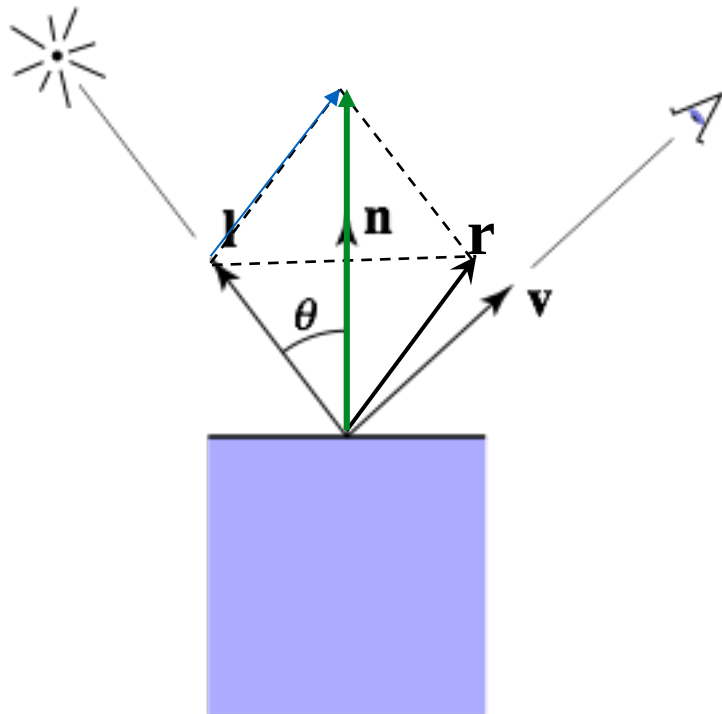
```
def shade(ray, lights, point, normal) :  
    result = ambient  
    for light in lights:  
        l=light.pos-position  
        shadowray=(point+e*normal,l)  
        if !scene.intersect(result,shadowray) :  
            it= surface.k*light.intensity*max(0,normal.l;  
            result+= surface.color*it  
    return result;
```



Mirror reflection

Intensity depends on view direction

– reflects incident light from mirror direction



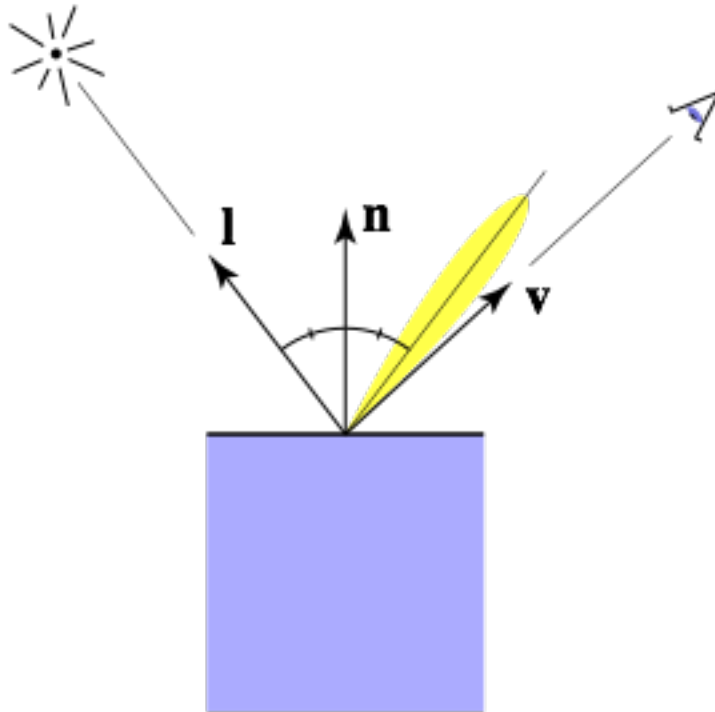
$$\mathbf{r} = 2(\mathbf{n} \cdot \mathbf{l})\mathbf{n} - \mathbf{l}$$

Specular shading (Phong)

Intensity depends on view direction

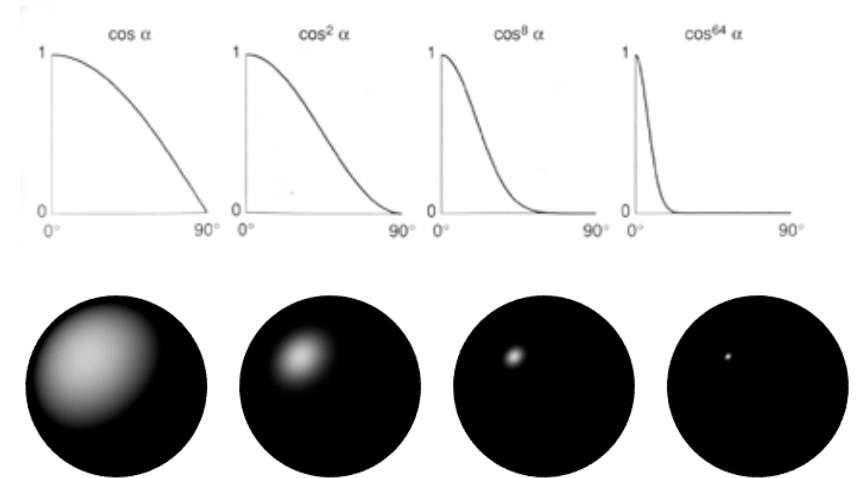
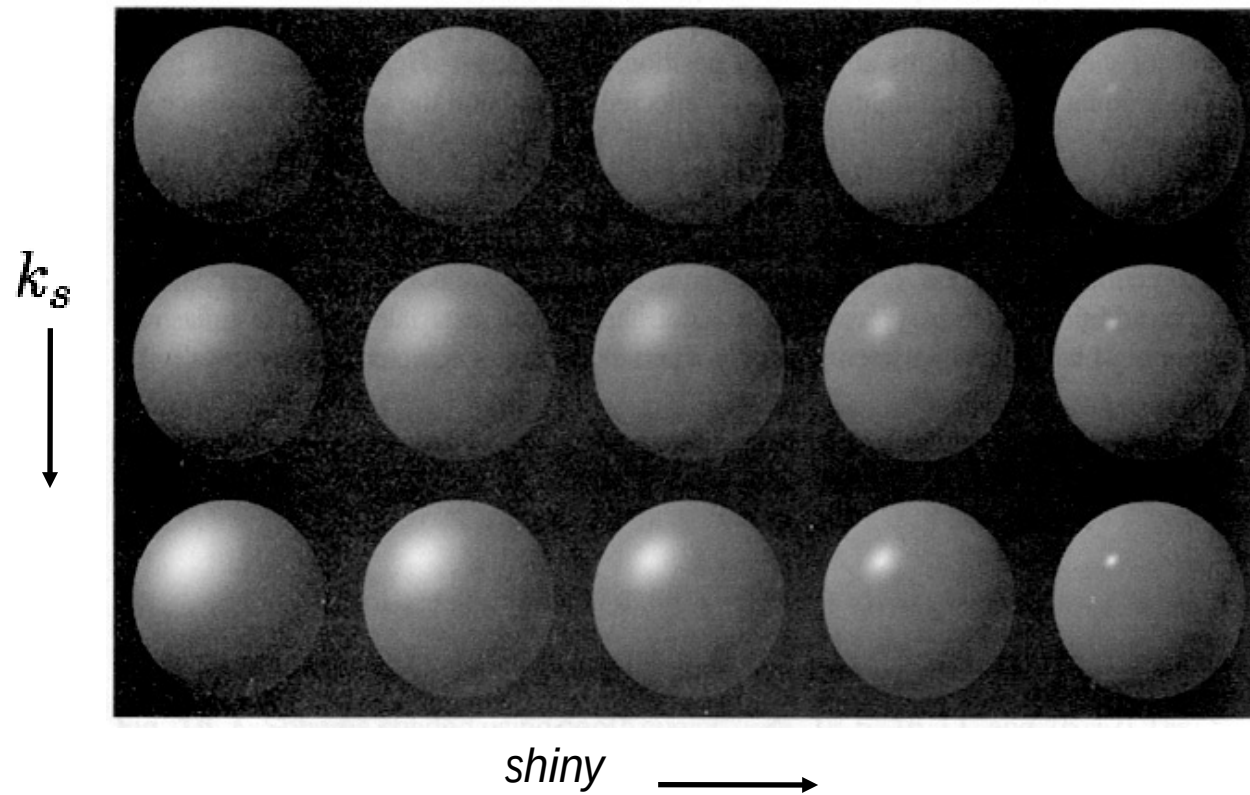
– bright near mirror configuration

$$k_s * I_s * (\mathbf{v} \cdot \mathbf{r})^{shiny}$$

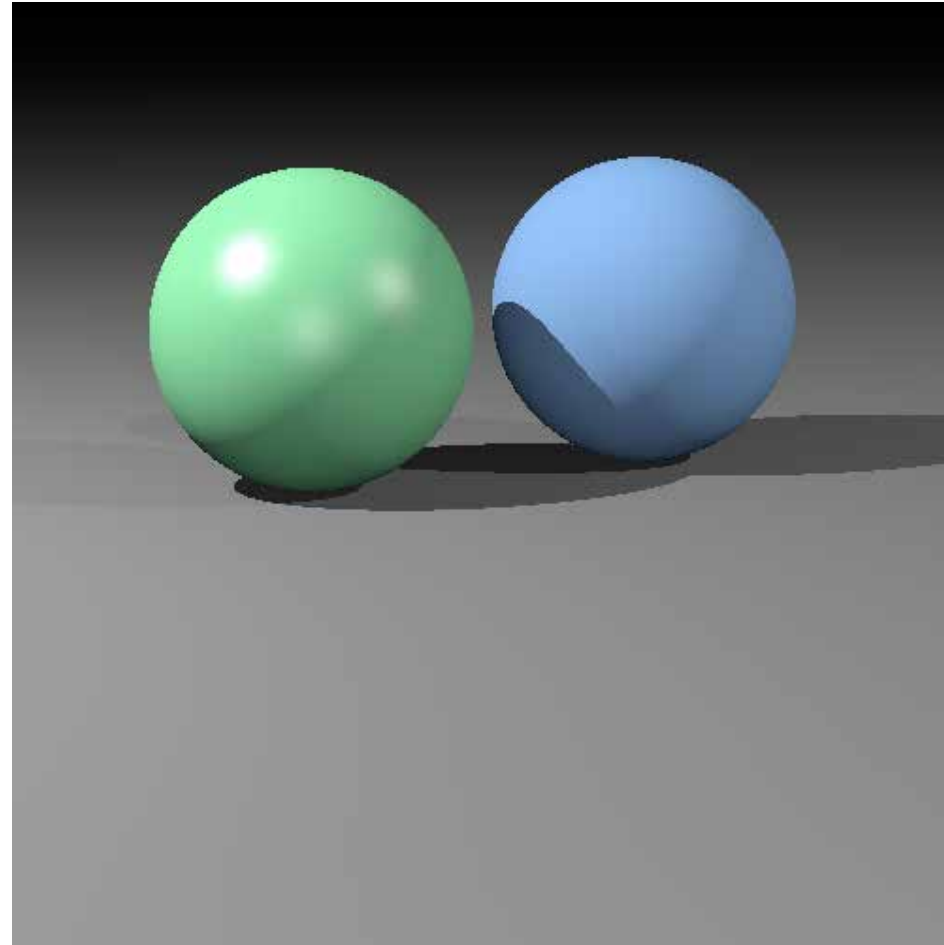


Phong model

Increasing *shiny* narrows the lobe

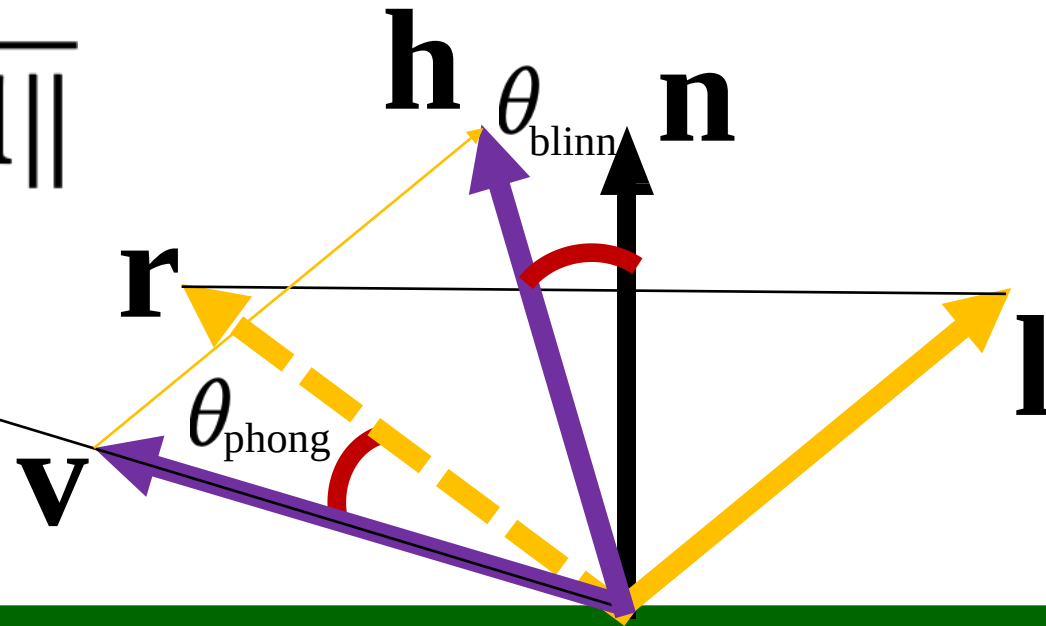
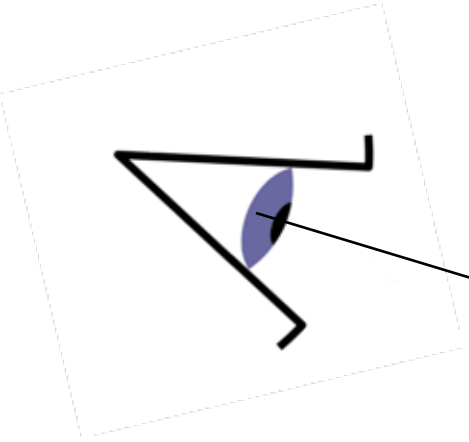


Diffuse + Specular (Phong) shading

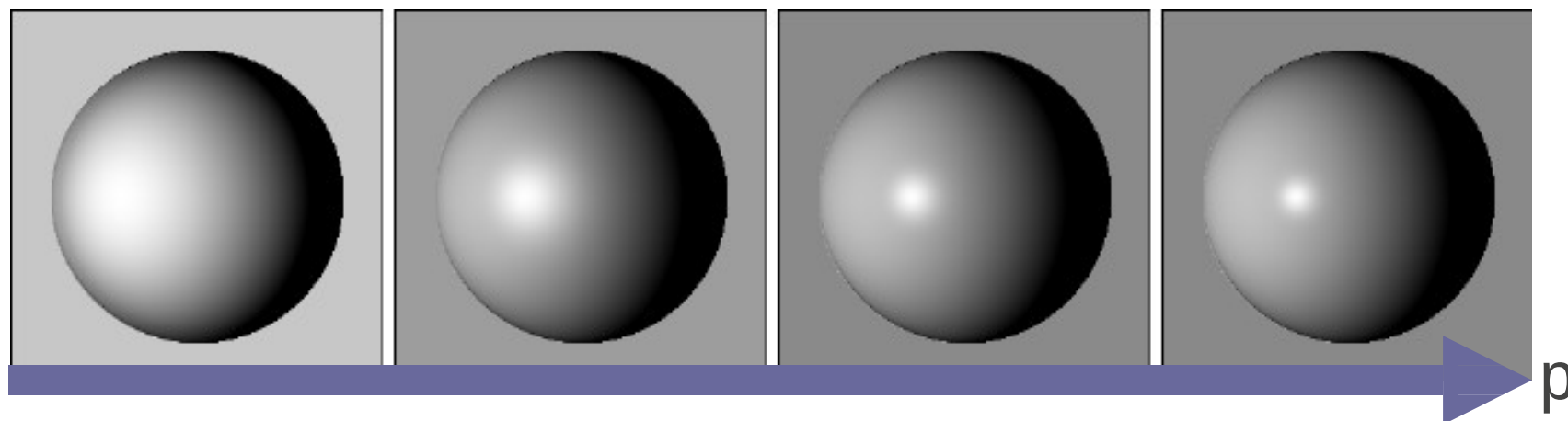
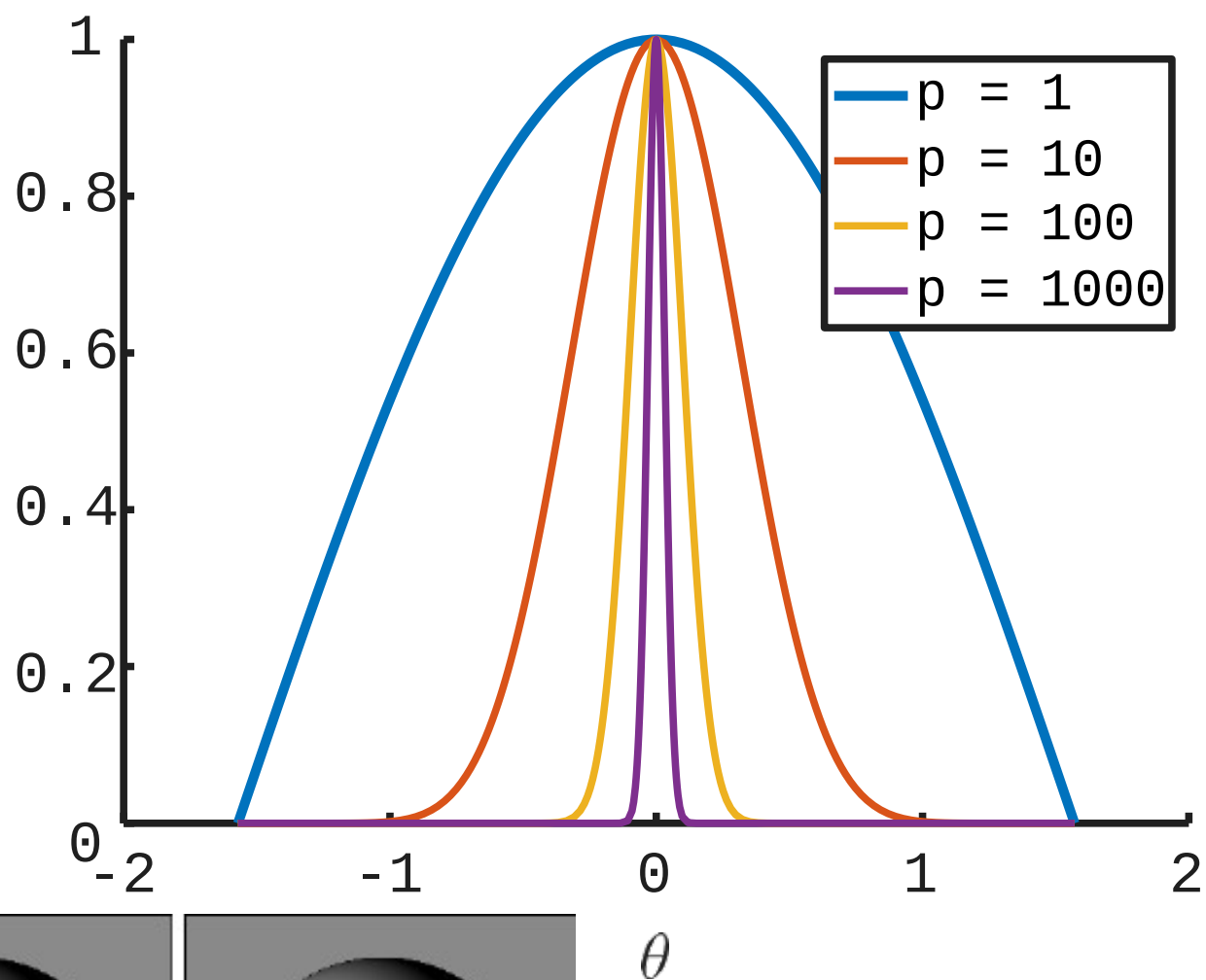


Specular Shading (Blinn)

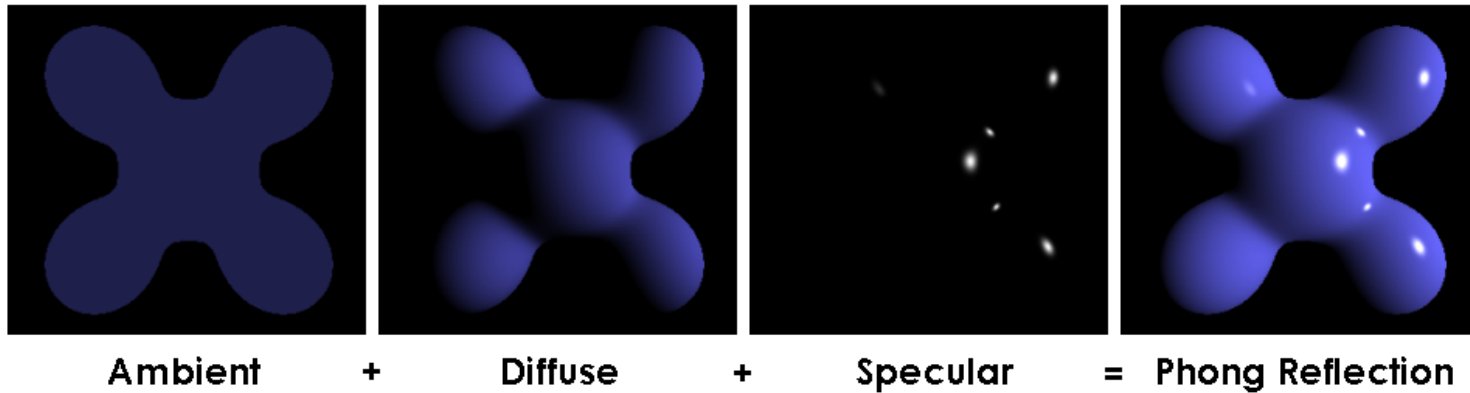
$$\mathbf{h} = \frac{\mathbf{v} + \mathbf{l}}{\|\mathbf{v} + \mathbf{l}\|}$$



$$L = k_s I \max(0, \mathbf{n} \cdot \mathbf{h})^p$$



Local Illumination



- Usually include ambient, diffuse, Phong in one model

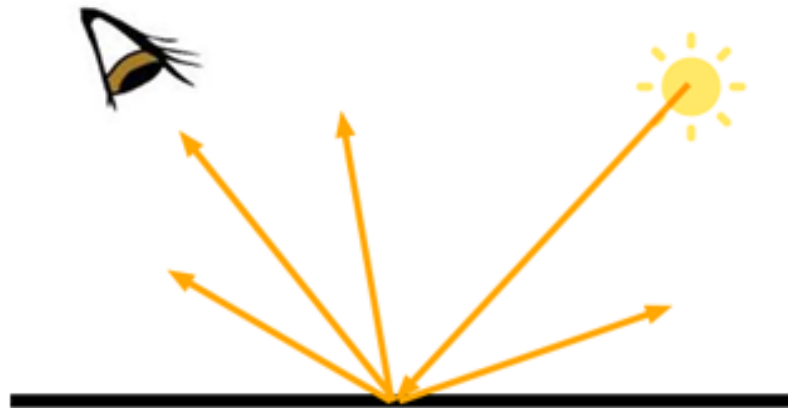
$$\begin{aligned} L &= L_a + L_d + L_s \\ &= k_a I_a + k_d (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{l}) + k_s (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{h})^p \end{aligned}$$

- The final result is the sum over many lights

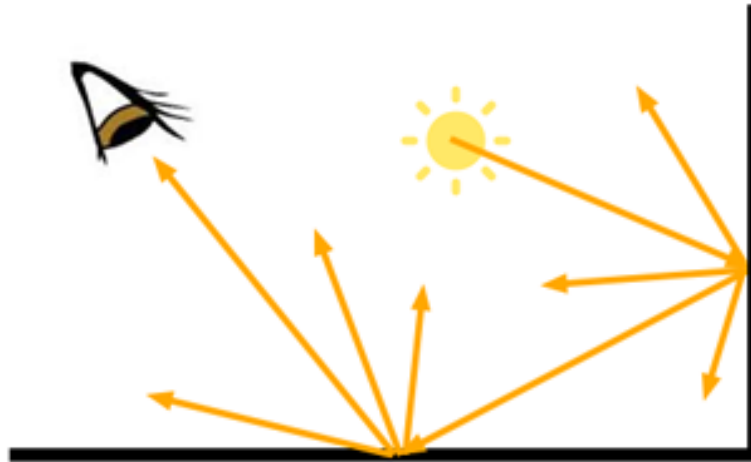
$$\begin{aligned} L &= L_a + \sum_{i=1}^N [(L_d)_i + (L_s)_i] \\ L &= k_a I_a + \sum_{i=1}^N \left[k_d (I_i/r_i^2) \max(0, \mathbf{n} \cdot \mathbf{l}_i) + \right. \\ &\quad \left. k_s (I_i/r_i^2) \max(0, \mathbf{n} \cdot \mathbf{h}_i)^p \right] \end{aligned}$$

Direct Illumination

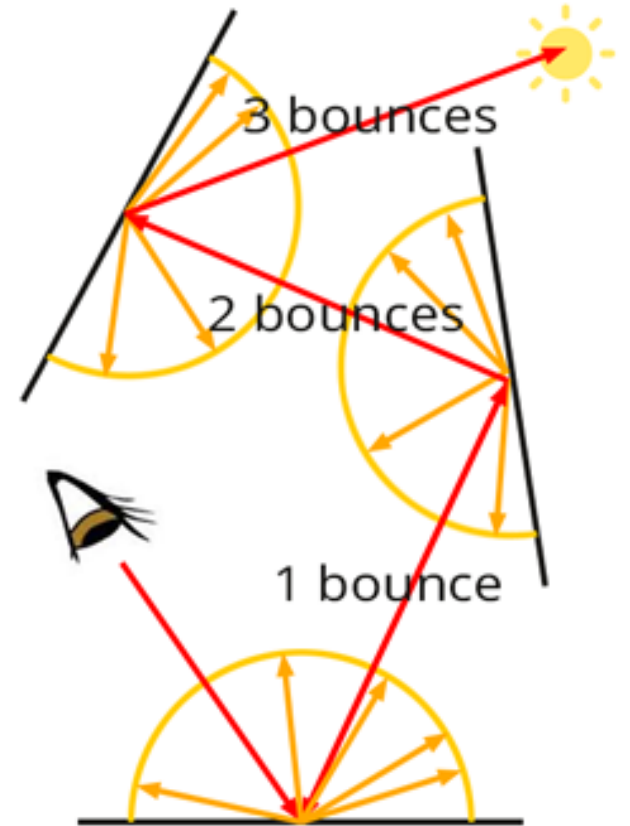
...no Global Effects so far



direct illumination

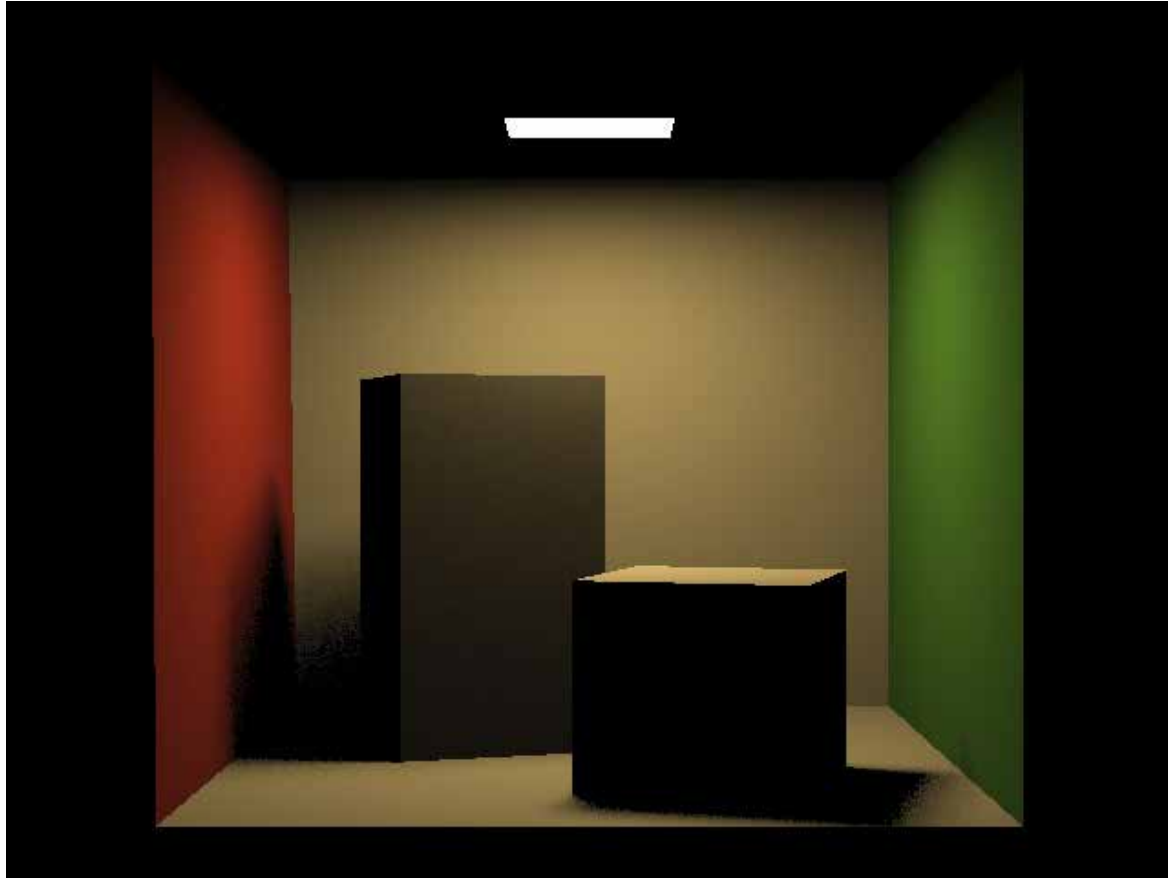


indirect illumination



© www.scratchapixel.com

Direct vs. Indirect Illumination



<http://www.deluxerender.com/2017/01/the-cornell-box-a-renderers-rite-of-pathage/>
https://en.wikipedia.org/wiki/Cornell_box

Local vs. Global Illumination

Local Illumination Models

- e.g. Phong, Blinn.
- Model source from a light reflected once off a surface towards the eye.
- Indirect light is included with an ad hoc “ambient” term which is normally constant across the scene.

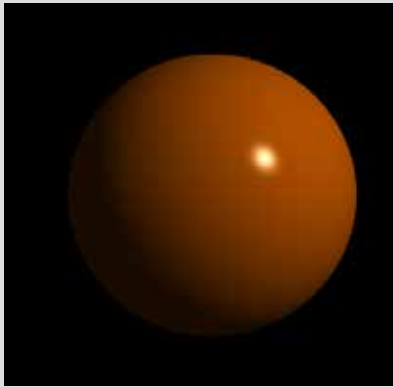
Global Illumination Models

- e.g. recursive ray tracing (incomplete model).
- Try to measure light propagation in the scene.
- Model interaction between objects, other objects, and their environment

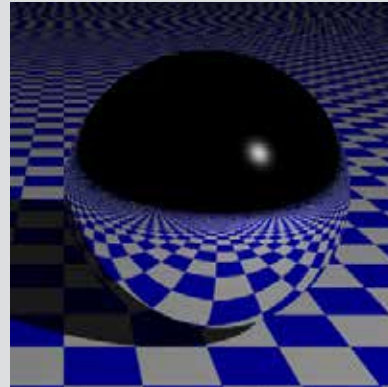
Path Tracing

- A ray from a light L can bounce of any number of specular S and diffuse objects D before entering the eye E. The paths from E to L for eg. can be
E-D-L, E-S-D-D-S-S-D-S-L, E-D-D-S-D-S-L...
- Rays are infinitely thin
- Don't disperse
- Ray Tracing model shiny objects exhibiting multiple reflections, i.e. paths of the form
E - S* - D⁺ - L.

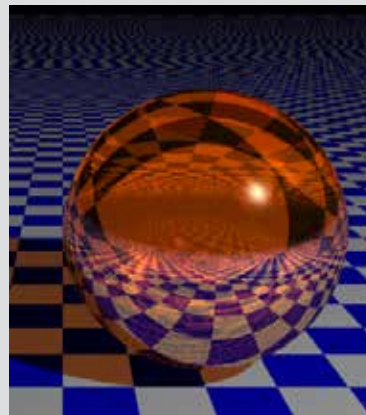
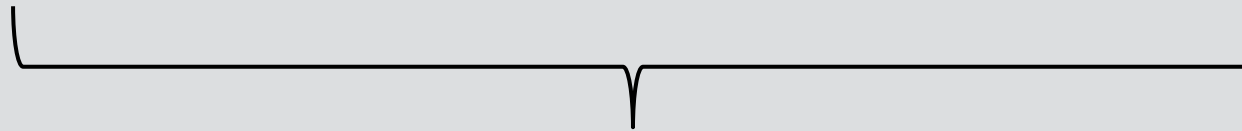
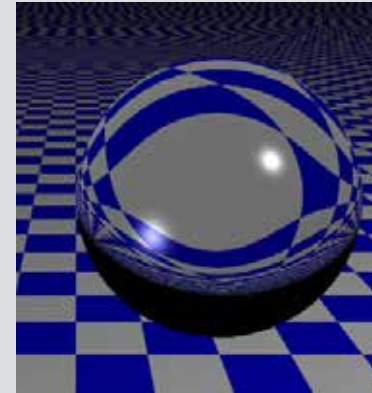
local illumination



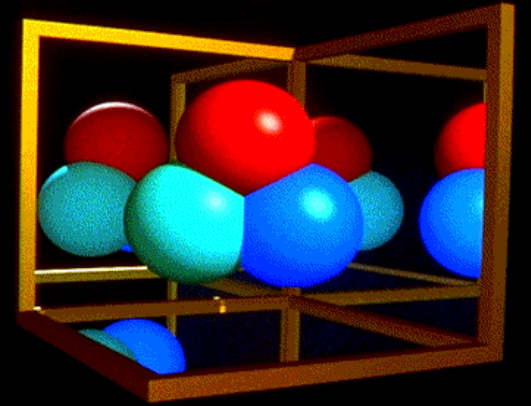
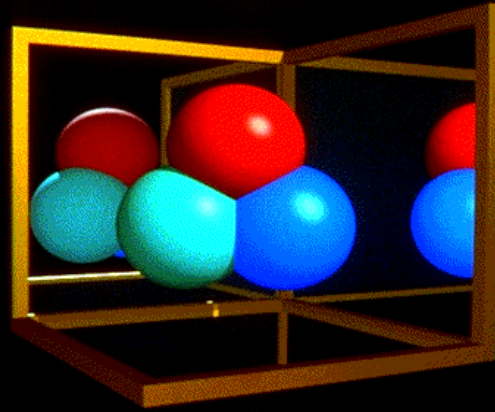
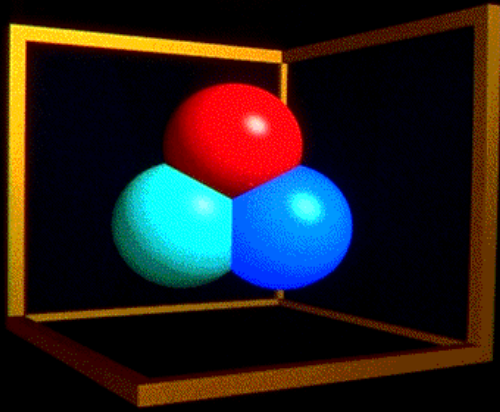
reflection



refraction

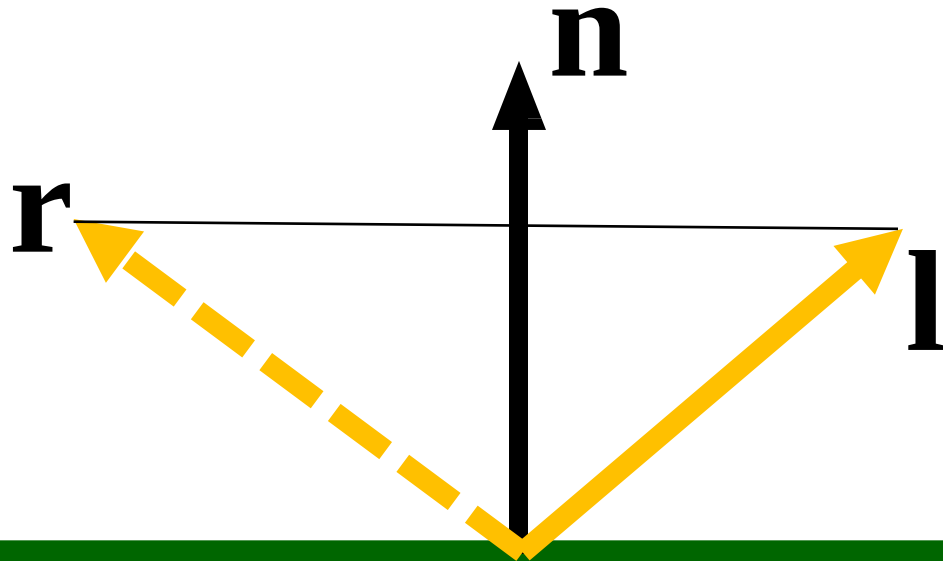


Ray Tracing recursion



Reminder: reflected ray

$$\mathbf{r} = 2(\mathbf{n} \cdot \mathbf{l})\mathbf{n} - \mathbf{l}$$



Ray Tracing

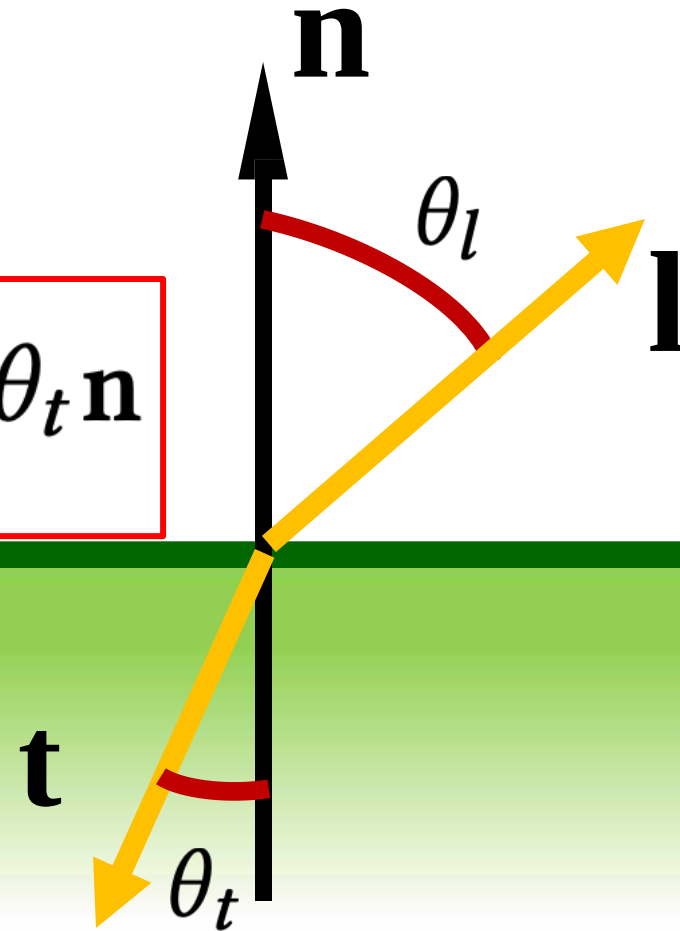
```
for each pixel in the image {  
    pixel colour = rayTrace(viewRay, 0)  
}
```

```
colour rayTrace(Ray, depth) {  
    for each object in the scene {  
        if(Intersect ray with object) {  
            colour = shading model  
            if(depth < maxDepth)  
                colour += rayTrace(reflectedRay, depth+1)  
        }  
    }  
    return colour  
}
```

Refraction (Snell's Law)

$$c_l \sin(\theta_l) = c_t \sin(\theta_t)$$

$$\mathbf{t} = -\frac{c_l}{c_t} \mathbf{l} + \frac{c_l}{c_t} \cos(\theta_l) \mathbf{n} - \cos \theta_t \mathbf{n}$$



Refraction (Snell's Law)

$$c_l \sin(\theta_l) = c_t \sin(\theta_t)$$

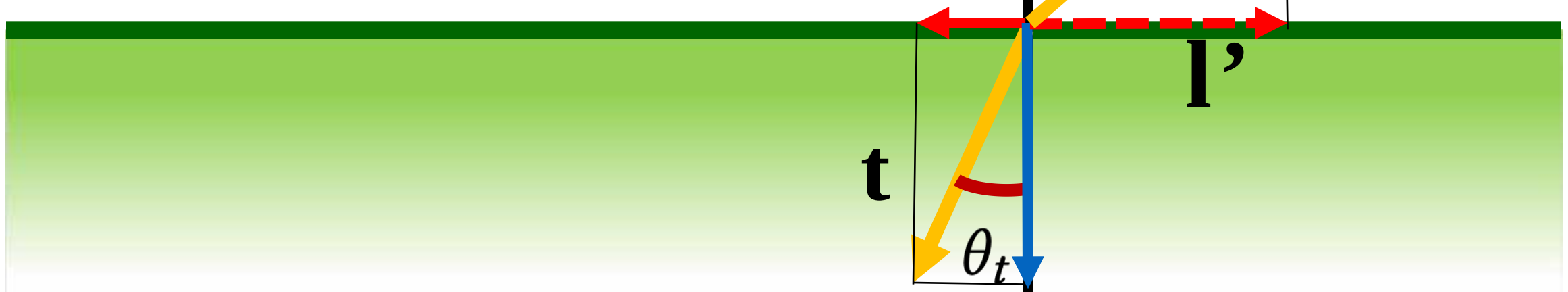
$$\mathbf{t} = -\frac{c_l}{c_t} \mathbf{l} + \frac{c_l}{c_t} \cos(\theta_l) \mathbf{n} - \cos \theta_t \mathbf{n}$$

$$\theta_t = \sin^{-1}(c_l/c_t \sin(\theta_l))$$

$$\mathbf{l} = \mathbf{l}' + \cos \theta_l \mathbf{n} \Rightarrow -\mathbf{l}' = -\mathbf{l} + \cos \theta_l \mathbf{n}$$

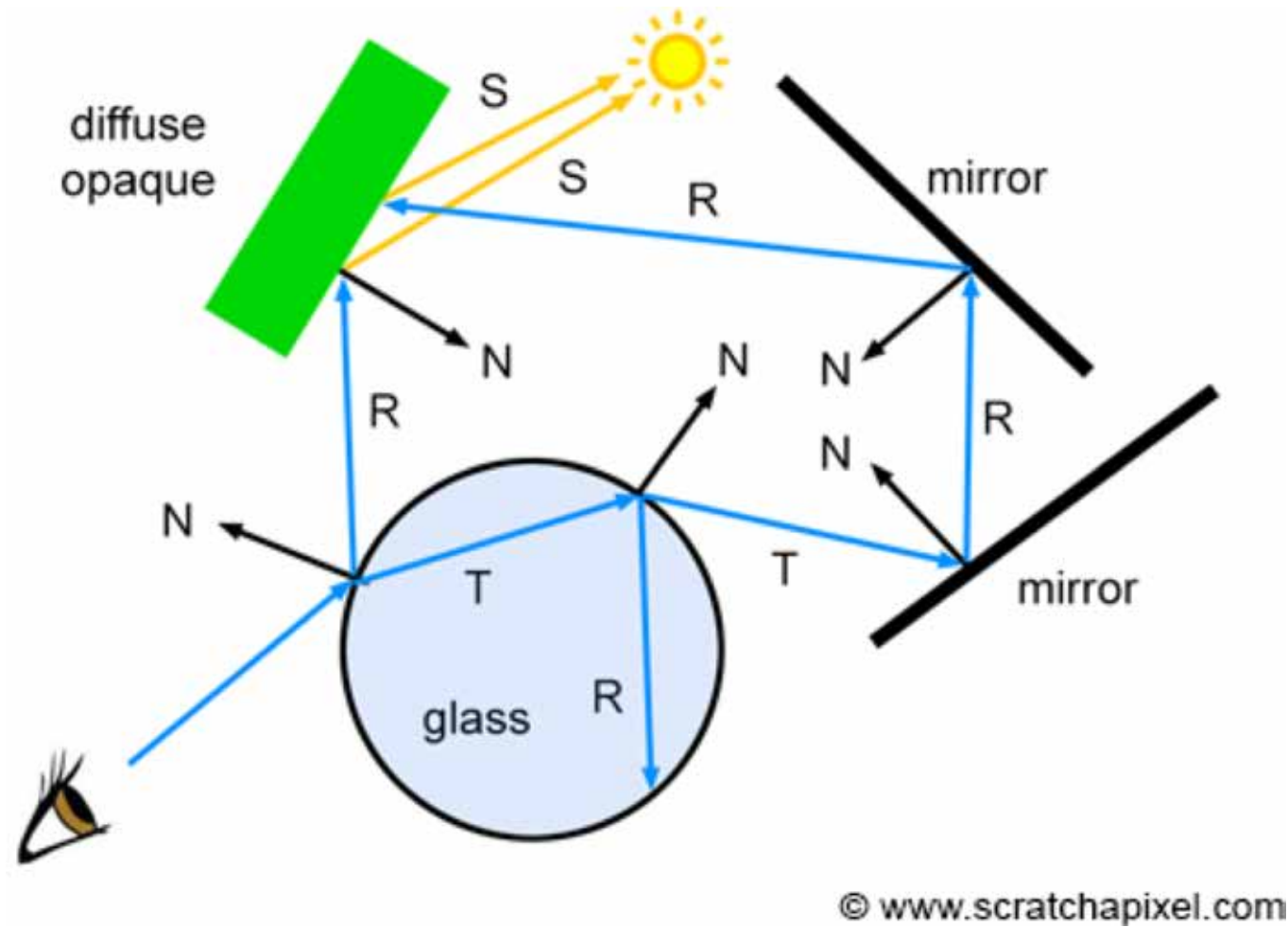
$$\mathbf{t}' = (||\mathbf{t}'||/||\mathbf{l}'||) * (-\mathbf{l} + \cos \theta_l \mathbf{n})$$

$$||\mathbf{t}'|| = \sin \theta_t, ||\mathbf{l}'|| = \sin \theta_l \Rightarrow ||\mathbf{t}'||/||\mathbf{l}'|| = c_l/c_t$$



```
def rayTrace(ray, depth) :  
    for each object in the scene :  
        if(intersect ray with object) :  
            colour = shading model  
            if(depth < maxDepth) :  
                colour += rayTrace(reflectedRay, depth+1)  
                colour += rayTrace(refractedRay, depth+1)  
    return colour
```

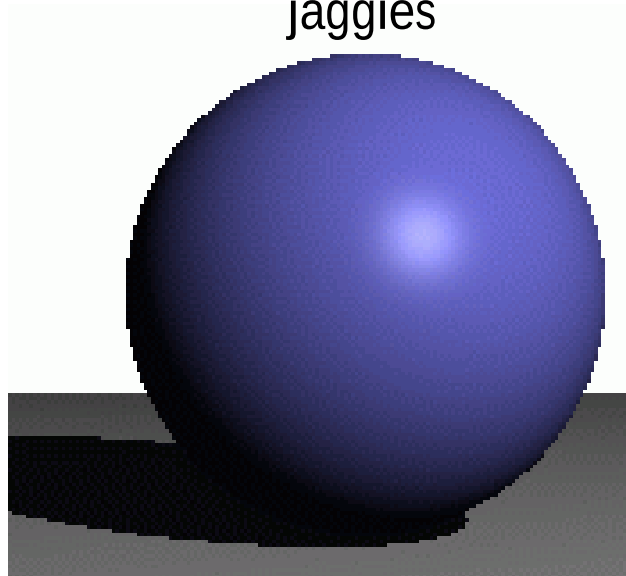
Ray Spawning



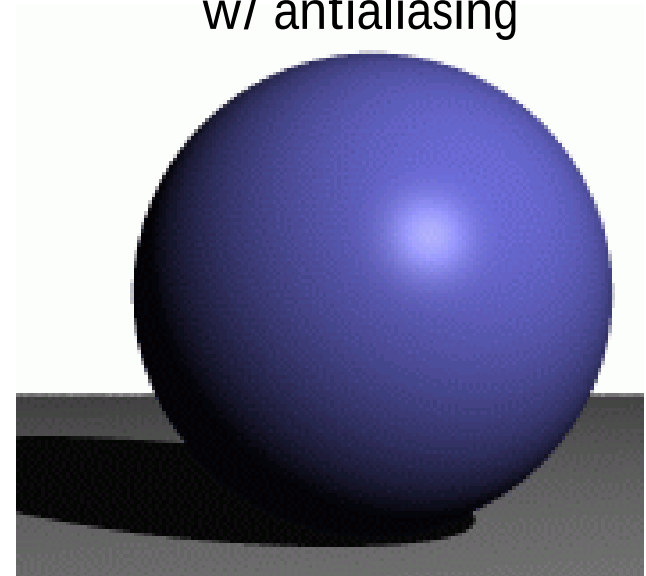
Ray Tracing supersampling

point light

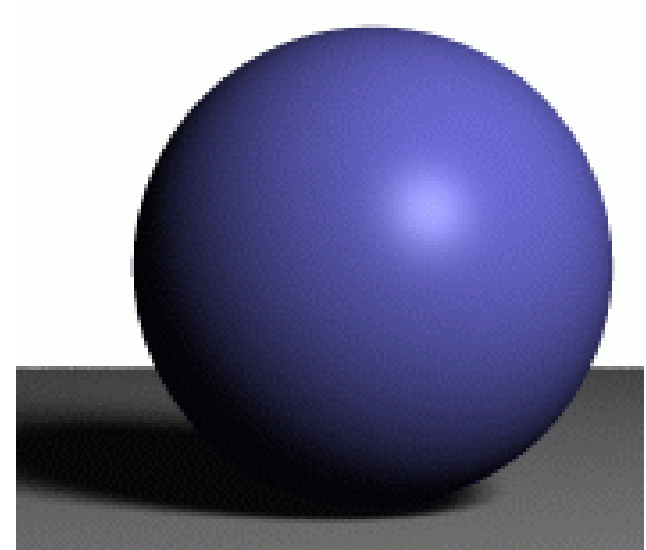
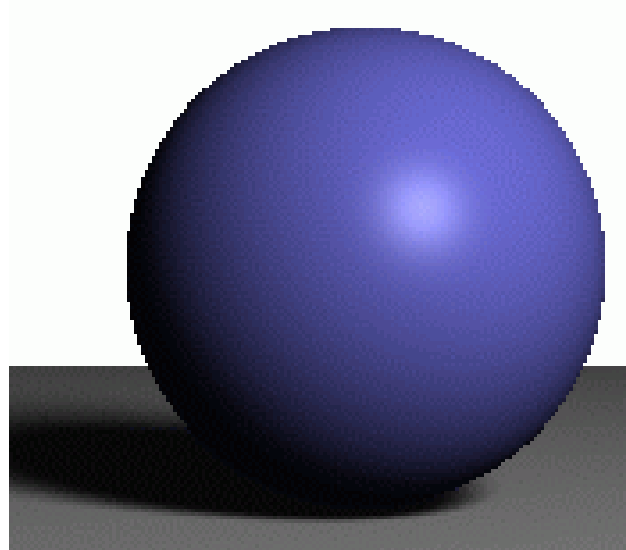
jaggies



w/ antialiasing



area light



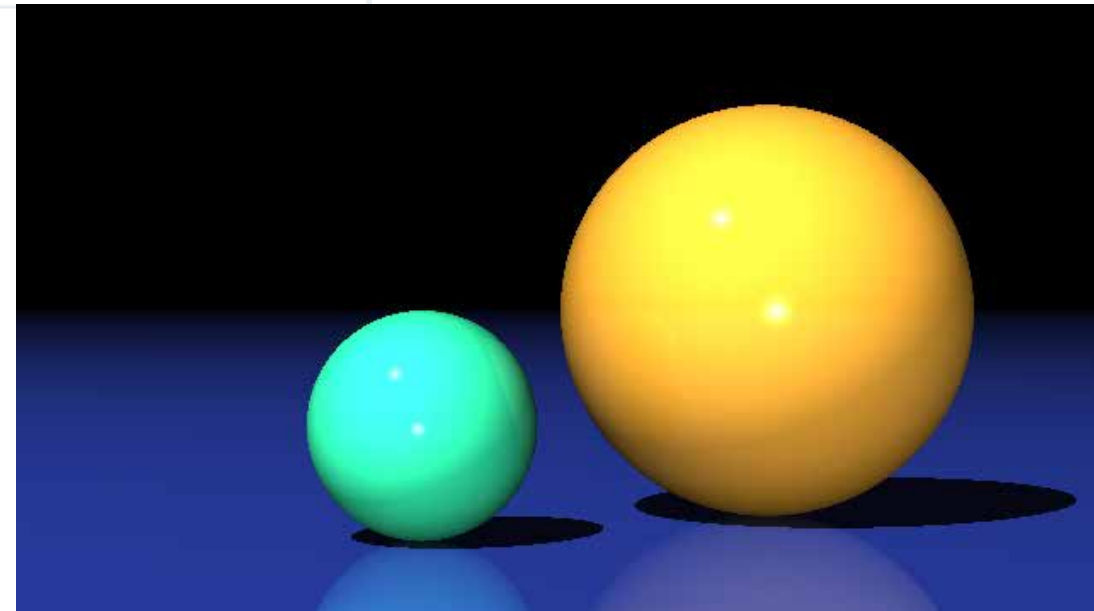
Ray Tracing

- Unifies in one framework
 - Hidden surface removal
 - Shadow computation
 - Reflection of light
 - Refraction of light
 - Global **specular** interaction

Assignment #2 tasks

Functions to implement (Part 2)

File	What it does
<code>src/reflect.py</code>	Mirror-reflect a direction about a normal.
<code>src/point_light_direction.py</code>	Direction and <code>max_t</code> toward a point light.
<code>src/directional_light_direction.py</code>	Direction and <code>max_t</code> toward a directional light.
<code>src/blinn_phong_shading.py</code>	Full Blinn-Phong colour with the per-light shadow test.
<code>src/raycolor.py</code>	Shoot a ray, shade the hit, and recurse for mirror reflections.



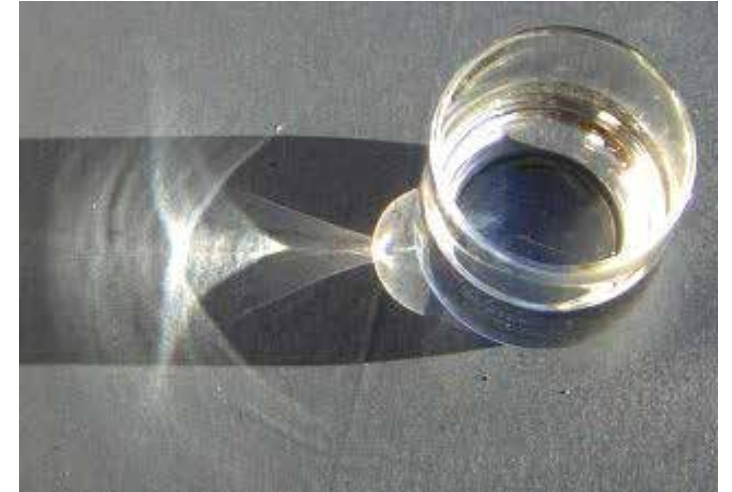
Ray Tracing Deficiencies

- Intersection computation time can be long (solution: **bounding volumes**).
- Recursive algorithm can lead to exponential complexity (solution: stochastic sampling).
- Ignores light transport mechanisms involving diffuse surfaces.

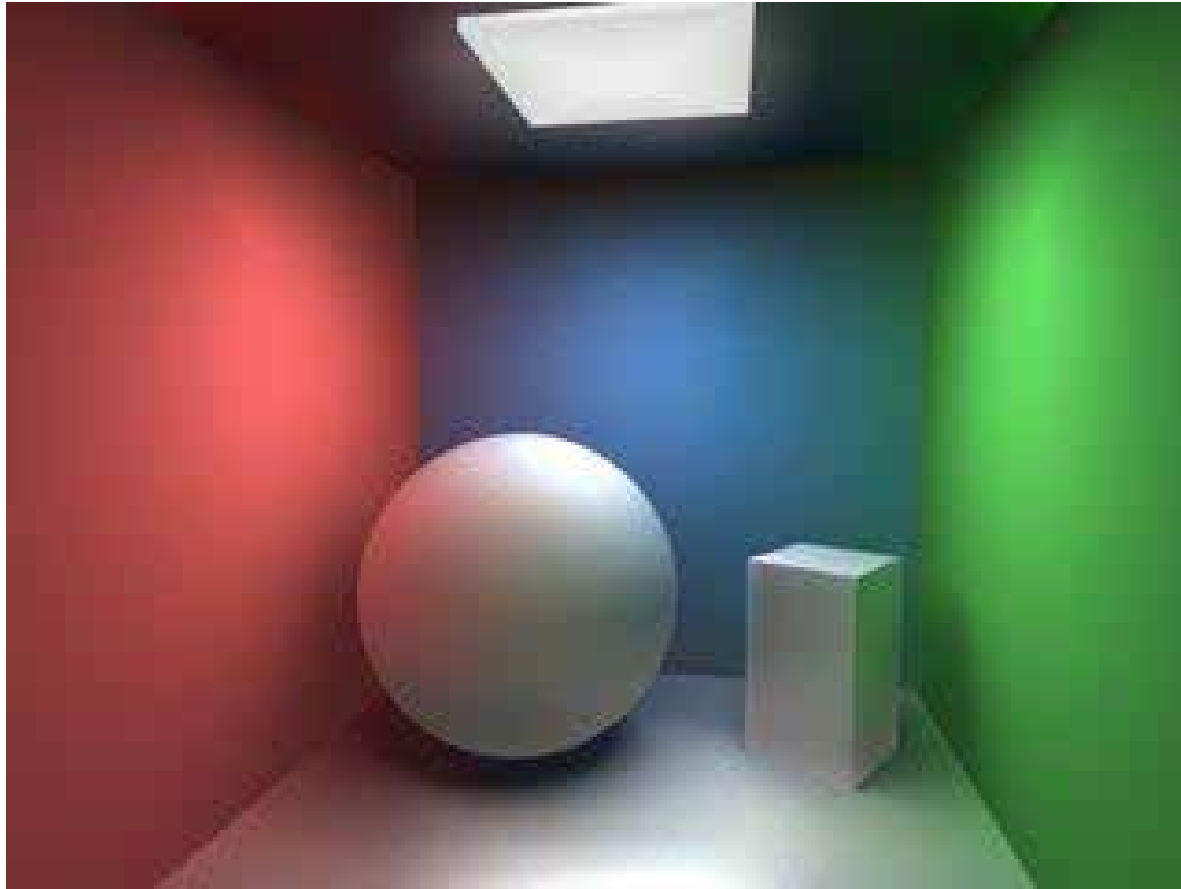


Ray Tracing Improvements: Caustics

- Transport $E - S - S - S - D - S - S - S - L$
- Trace from the light to the surfaces and then from the eye to the surfaces
- “shower” scene with light and then collect it
- “Where does light go?” vs “Where does light come from?”
- Good for caustics



Radiosity: E – D – D – D - L



The Rendering Equation



$$L_o(x, \vec{w}) = L_e(x, \vec{w}) + \int_{\Omega} f_r(x, \vec{w}', \vec{w}) L_i(x, \vec{w}') (\vec{w}' \cdot \vec{n}) d\vec{w}'$$

scarier version found here: https://en.wikipedia.org/wiki/Rendering_equation

research paper found here: <https://dl.acm.org/doi/10.1145/15886.15902>

The Rendering Equation

$$L_o(x, \vec{w}) = L_e(x, \vec{w}) + \int_{\Omega} f_r(x, \vec{w}', \vec{w}) L_i(x, \vec{w}') (\vec{w}' \cdot \vec{n}) d\vec{w}'$$

outgoing light at position $\mathbf{x}$ and
direction $\mathbf{w}$

emitted light at position $\mathbf{x}$ and
direction $\mathbf{w}$

and

reflected light at position $\mathbf{x}$ and
direction $\mathbf{w}$

The Rendering Equation

$$L_o(x, \vec{w}) = L_e(x, \vec{w}) + \int_{\Omega} f_r(x, \vec{w}', \vec{w}) L_i(x, \vec{w}') (\vec{w}' \cdot \vec{n}) d\vec{w}'$$

the reflected light at position $\mathbf{x}$
and direction $\mathbf{w}$

is the integral over all
possible directions $\mathbf{w}'$

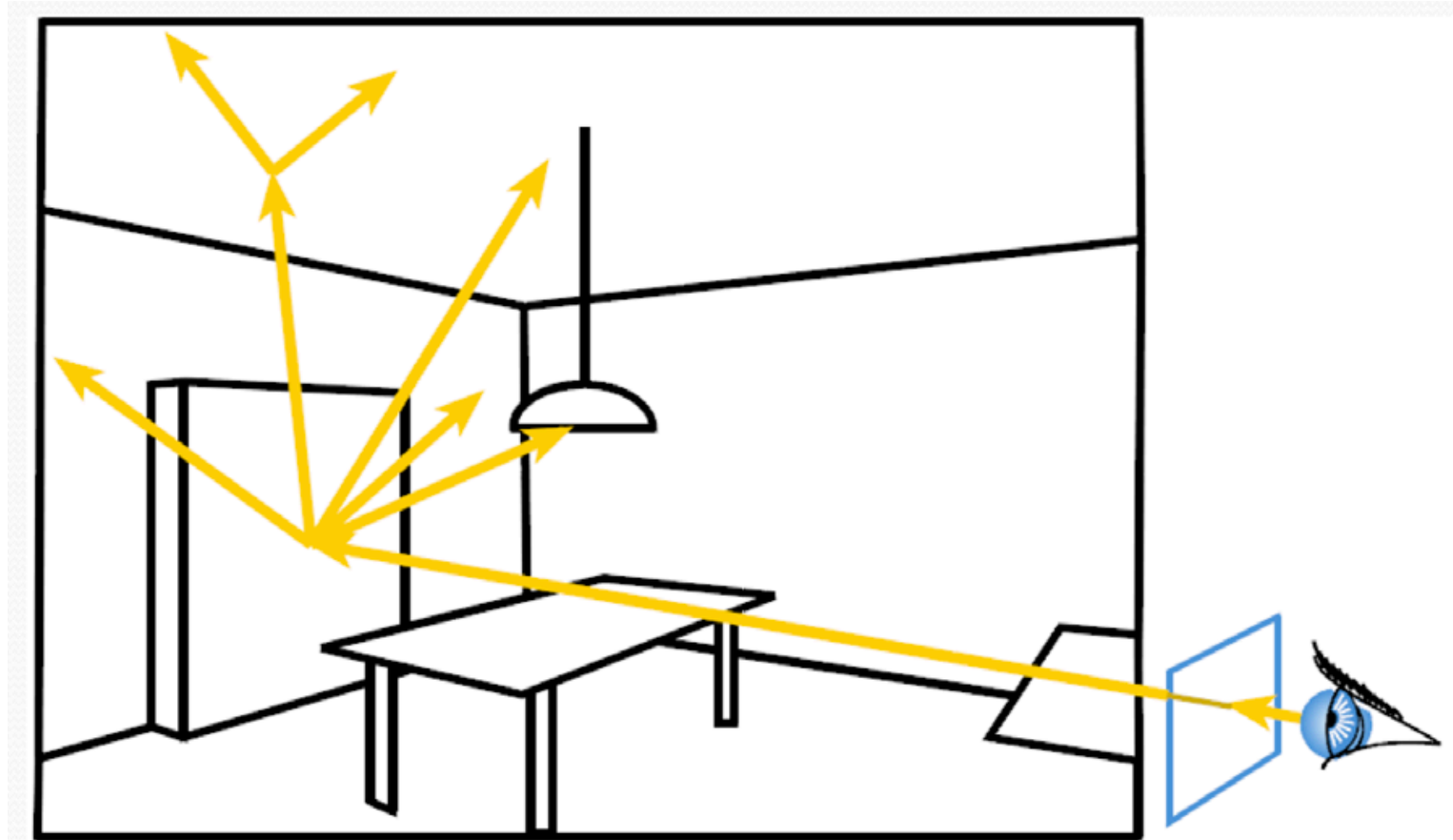
the incoming light from all
directions

times

BRDF:
a function describing
how light is reflected at
an opaque surface

Monte Carlo Methods

Rely on random sampling to “solve” rendering equation



Area Light

Hard v soft shadows



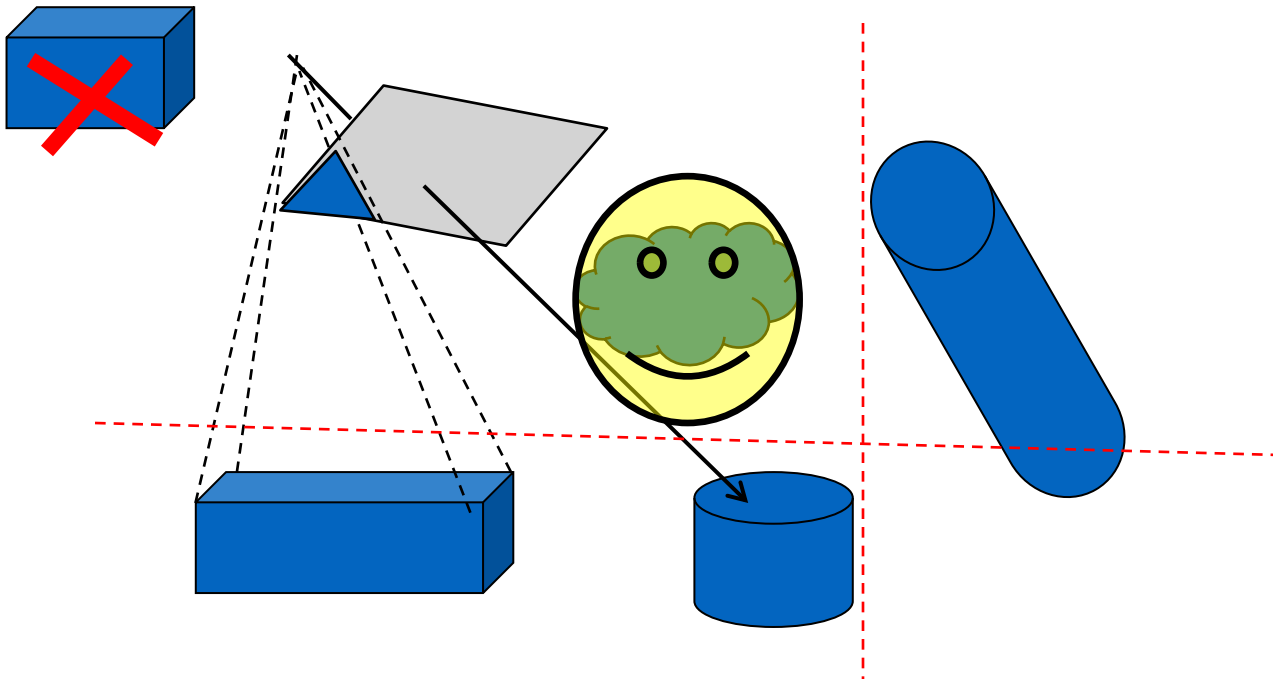
Hard shadow

More realistic soft shadows

Ray Intersection: Efficiency Considerations

Speed-up the intersection process.

- Ignore object that clearly don't intersect.
- Use proxy geometry.
- Subdivide and structure space hierarchically.
- Project volume onto image to ignore entire sets of rays.



Faster Intersections for Ray Tracing

